

Activation of Polycystin-1 Signaling by Binding of Stalk-derived Peptide Agonists

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Abstract

Polycystin-1 (PC1) is the membrane protein product of the PKD1 gene whose mutation is responsible for 85% of the cases of autosomal dominant polycystic kidney disease (ADPKD). ADPKD is primarily characterized by the formation of renal cysts and potential kidney failure. PC1 is an atypical G protein-coupled receptor (GPCR) consisting of 11 transmembrane helices and an autocatalytic GAIN domain that cleaves PC1 into extracellular N-terminal (NTF) and membrane-embedded C-terminal (CTF) fragments. Recently, signaling activation of the PC1 CTF was shown to be regulated by a stalk tethered agonist (TA), a distinct mechanism observed in the adhesion GPCR family. A novel allosteric activation pathway was elucidated for the PC1 CTF through a combination of Gaussian accelerated molecular dynamics (GaMD), mutagenesis and cellular signaling experiments. Here, we show that synthetic, soluble peptides with 7 to 21 residues derived from the stalk TA, in particular, peptides including the first 9 residues (p9), 17 residues (p17) and 21 residues (p21) exhibited the ability to re-activate signaling by a stalkless PC1 CTF mutant in cellular assays. To reveal molecular mechanisms of stalk peptide-mediated signaling activation, we have applied a novel Peptide GaMD (Pep-GaMD) algorithm to elucidate binding conformations of selected stalk peptide agonists p9, p17 and p21 to the stalkless PC1 CTF. The simulations revealed multiple specific binding regions of the stalk peptide agonists to the PC1 protein including an “intermediate” bound yet inactive state. Our Pep-GaMD simulation findings were consistent with the cellular assay experimental data. Binding of peptide agonists to the TOP domain of PC1 induced close TOP-putative pore loop interactions, a characteristic feature of the PC1 CTF signaling activation mechanism. Using sequence covariation analysis of PC1 homologs, we further showed that the peptide binding regions were consistent with covarying residue pairs identified between the TOP domain and the stalk TA. Therefore, structural dynamic insights into the mechanisms of PC1 activation by stalk-derived peptide agonists have enabled an in-depth understanding of PC1 signaling. They will form a foundation for development of PC1 as a therapeutic target for the treatment of ADPKD.

eLife assessment

This joint computational/experimental study demonstrates the ability of synthetic peptides derived from the stalk-tethered agonist in Polycystin-1 (PC1) to re-activate signaling by a stalkless C-terminal fragment of PC1. The study is **valuable** as it discovered peptide agonists for PC1 and the integrated *in vitro* and *in silico* approach is potentially applicable to the analysis of related systems. Following the revision, the line of evidence presented in the current manuscript is considered **convincing**.

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Introduction

Polycystin-1 (PC1) is the protein product of the PKD1 gene that is mutated in the majority of cases (~85%) of autosomal dominant polycystic kidney disease (ADPKD)¹. ADPKD is a potentially lethal disease, affecting >0.6 million individuals in the US. It causes renal cyst formation that could consequently lead to kidney failure. Currently, the only approved treatment for ADPKD is JynarqueTM, a small-molecule antagonist of the arginine vasopressin receptor 2, V2R, whose signaling, and production of cAMP has been shown to be increased in PKD. This drug targets one of the aberrant pathways downstream from the PKD gene mutation but is inadequate due to its limitations in only slowing disease progression and causing adverse side effects². ADPKD severity is dependent on the functional level of PC1, and as such, therapies designed to increase the level of PC1 protein, and its functionality are currently being pursued^{3–7}. Approximately one-third of PKD1 mutations are non-truncating and could encode partially functional PC1 protein^{3, 8–10}. As such, therapeutic treatments that directly target and activate PC1 may represent a promising approach for the treatment of ADPKD. However, this approach remains difficult due to incomplete knowledge of the proximal-most functions of PC1.

PC1 shares characteristics with the Adhesion class of GPCRs (ADGRs), including a conserved GPCR autoproteolysis inducing (GAIN) domain that directs autocatalytic cleavage at an embedded GPCR proteolysis site (GPS) motif¹¹. Intramolecular cleavage at the GPS motif generates two non-covalently attached fragments - the extracellular N-terminal fragment (NTF) and the membrane-embedded C-terminal fragment (CTF)^{12, 13}. Similar to ADGRs, the PC1 NTF consists of multiple adhesive domains that promote interactions between cells and with the extracellular matrix^{14–18}, while the PC1 CTF is composed of 11 transmembrane (TM) helices and a short C-terminal tail (C-tail)¹⁹ that has been shown to interact with G proteins for signaling activation or regulation^{20, 21} and has thus led to description of PC1 as an atypical GPCR. Previous studies demonstrated the critical importance of cleavage at the PC1 GPS site to prevent renal cystogenesis in mouse models^{4, 22}. For the ADGRs, a tethered agonist (TA) model has been proposed for activation of G protein signaling. After dissociation of the NTF, the N-terminal stalk of the ADGR CTF interacts with its membrane-embedded TM domains to induce conformational rearrangements that mediate activation of G protein signaling^{23–26}. Exogenous synthetic peptides consisting of various lengths of the N-terminal sequence of the stalk have been shown to function as soluble agonists in activation of signaling by full-length and CTF mutants for numerous ADGRs^{20, 27}.

In previous studies of the PC1 CTF, we revealed a stalk TA-dependent molecular mechanism underlying CTF-mediated activation of an NFAT promoter luciferase reporter through complementary *in vitro* cell signaling experiments and all-atom Gaussian accelerated Molecular Dynamics (GaMD) simulations²⁸. GaMD is an unconstrained enhanced sampling method that

works by adding a harmonic boost potential to reduce large biomolecular energy barriers²⁹ and has been used successfully to capture multiple complex biological processes^{30–37,38–44,30,32} including GPCR activation³¹. Expression constructs encoding a stalkless PC1 CTF (a nonbiological mutant with deletion of the first 21 N-terminal residues of CTF) and three ADPKD-associated missense mutants within the stalk region (G3052R, R3063C, and R3063P) were shown to be defective in reporter activation as compared to wildtype PC1 CTF. GaMD simulations revealed a novel allosteric transduction pathway for activation of PC1 CTF signaling that involves initiation by the Stalk interacting with a large extracellular loop between TM segments S1/TM6 and S2/TM7, called the TOP domain, followed by close interactions between the TOP and a putative pore loop (PL) domain between the final 2 TM domains. GaMD simulations of the wildtype PC1 CTF also identified a “Closed/Active” low-energy state related to the large number of Stalk-TOP contacts and the R3848-E4078 ionic interaction between the TOP and PL domains that was not present in the stalkless CTF²⁸.

Here, we have utilized *in vitro* cell signaling assays to identify peptide agonists targeting PC1 in combination with *in silico* studies to investigate their binding mechanisms for activation of PC1 signaling. Synthetic peptides of 7-21 residues in length derived from the N-terminus of the PC1 CTF stalk sequence were tested for their ability to re-activate signaling of the stalkless CTF expression construct. Peptide docking and simulations with the recently developed Peptide GaMD (Pep-GaMD), which is able to characterize peptide-protein binding processes more efficiently⁴⁵, were combined for selected peptide agonists p9, p17 and p21 to gain insight into their binding mechanism to the stalkless PC1 CTF. Pep-GaMD was able to successfully refine the docking conformations of the peptides bound to the extracellular TOP domain of PC1. In further Pep-GaMD simulations, the key salt bridge interaction between R3848 and E4078 from the TOP domain and PL, respectively, was observed upon binding of the peptides to stalkless PC1 CTF. Using Potts covariation analysis, in which a protein fitness model is inferred based on observed mutational covariation patterns in multiple sequence alignments (MSAs) of homologous proteins⁴⁶, we identified residues in the PC1 stalk with direct mutational covariation with residues in the TOP domain, which were strikingly consistent with the binding interfaces identified in docking and simulation studies. Overall, these analyses yielded mechanistic insights underlying the stalk peptide agonist mediated signal re-activation of stalkless PC1 CTF. Such insights provide significant contributions toward the future design and development of peptide modulators targeting PC1 for an effective ADPKD therapeutic treatment.

Results

Synthetic, stalk-derived peptides re-activate NFAT reporter by CTF^{Δst} *in trans*

Our previous study utilized expression constructs of human PC1 CTF. However, in order to prepare for eventual *in vivo* experiments in mouse models, we generated expression constructs of mouse (m) PC1 consisting of the signal peptide sequence of the T cell surface glycoprotein CD5 (MPMGSLLQPLATLYLLGMLVASVLG) fused in frame with the stalk sequence of wildtype mCTF beginning with residue T3041, or with a ‘stalkless’ CTF lacking the first 21 residues of the stalk (mCTF^{Δst}) beginning with residue S3062 (**Fig. 1A**). The CD5 signal peptide coding sequence was added to the wildtype mCTF and stalkless mCTF^{Δst} expression constructs in order to ensure their translation at the endoplasmic reticulum for plasma membrane localization.

Transient transfection of HEK293T cells with either empty expression vector (ev), CTF or CTF^{Δst} showed that the CTF^{Δst} mutant exhibited a dramatic loss of NFAT reporter activation that was essentially reduced to ev control levels (**Fig. 1B**). Both total (**Fig. 1C-D**) and cell surface (**Fig. S1A-B**) expression levels of CTF^{Δst} were comparable to CTF, which suggests that neither protein

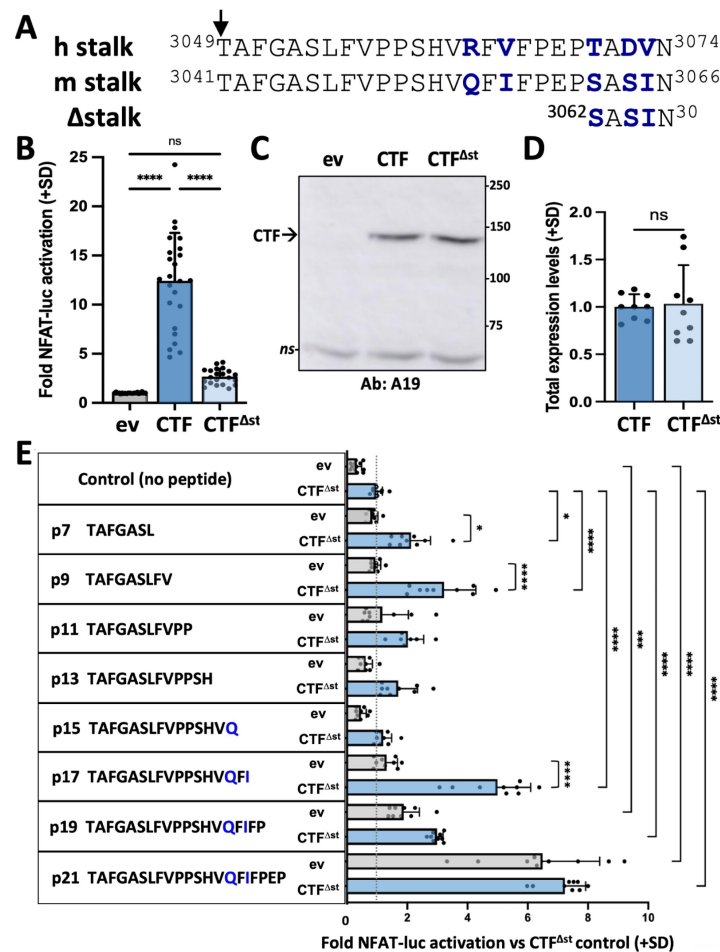


Figure 1.

Synthetic peptides derived from the stalk sequence of PC1 can stimulate signaling of stalkless PC1 CTF.

(A) Alignment of CTF stalk sequences from human (h) and mouse (m) PC1. CTF^{Δst} has a 21-residue deletion from the N-terminal end of the stalk region. Arrow, GPS cleavage site. Non-identical residues shown in bolded blue. (B) Activation of the NFAT-luc reporter by transfected mCTF or mCTF^{Δst} expression constructs shown relative to empty expression vector (ev) as means (+ standard deviation, SD) of 3 wells/construct from each of 7 independent experiments. (C) Representative Western blot of total cell lysates from one of the experiments in (B), probed with antisera A19 against mouse PC1 C-tail. ns, non-specific. (D) Summary of the total expression levels (means +SD) of CTF^{Δst} relative to CTF from the experiments in (B). (E) Stalk peptide treatment of ev- or mCTF^{Δst}-transfected cells. Sequences of stalk-derived peptides p7-p21 are shown. Graph represents the fold NFAT-luc activation for both ev- (gray bars) and CTF^{Δst}- (blue bars) transfected cells relative to the CTF^{Δst} control after 24 hr treatment with or without peptide. Results are the means (+SD) of 3 separate experiments, each with 3 wells/condition. *, $p < 0.05$; ***, $p = 0.0001$; ****, $p < 0.0001$. Analysis by 1-way ANOVA with Tukey-Kramer post-test.

stability nor membrane trafficking was responsible for the inability of CTF^{Δst} to activate the NFAT reporter. These results are consistent with those obtained using expression constructs of human PC1 that demonstrated the stalk region of PC1 CTF acts as a tethered peptide agonist²⁸.

To further investigate the agonistic property of the CTF stalk, we synthesized peptides (p) consisting of the N-terminal 7, 9, 11, 13, 15, 17, 19 or 21 residues from the stalk sequence of mPC1. All peptides were appended with a C-terminal, 7-residue hydrophilic sequence (GGKKKKK) to increase solubility. HEK293T cells were transiently transfected with empty expression vector or mCTF^{Δst} plasmids along with the NFAT luciferase reporter and then treated with stalk peptides p7 through p21 or with addition of culture medium only ('no peptide' control).

The NFAT reporter was significantly activated in CTF^{Δst}-transfected cells by treatment with p7, p9 or p17 as compared to their corresponding ev + peptide treatment controls. These stalk peptides also significantly increased reporter activity in comparison to the CTF^{Δst} with no peptide treatment control (**Fig. 1E**). Treatment of CTF^{Δst}-transfected cells with p19 or p21 also significantly increased reporter activation in comparison to the CTF^{Δst} + no peptide control; however, reporter activation occurred in both ev- and CTF^{Δst}-transfected cells treated with either p19 or p21, suggesting that p19- and p21-mediated activation was not dependent on exogenous expression of mouse CTF^{Δst} and could be activating the endogenous human PC1 protein. Treatment of ev- and CTF^{Δst}-transfected cells with peptide consisting of the hydrophilic sequence alone (i.e., solubility peptide) showed that the solubility tag was not responsible for the rescue of CTF^{Δst}-mediated reporter activation by stalk peptides such as p17 (**Fig. S2**). Altogether, these results were consistent with soluble stalk-derived peptides acting as PC1 CTF agonists *in trans*, and provided additional support for the PC1 CTF stalk region harboring tethered agonist (TA) activity²⁸. We hypothesized that the soluble, activating peptides bind to the TOP domain of PC1 in a manner mimicking the tethered stalk in order to reactivate signaling of the stalkless CTF^{Δst} mutant. From among the active stalk-derived peptides, we selected p9, p17 and p21 that exhibited the highest agonist activity in CTF^{Δst} reporter activation (**Fig. 1E**) for *in silico* simulation studies.

Docking and Pep-GaMD simulations of peptide agonist binding to stalkless PC1 CTF

We chose to use the stalkless CTF (ΔStalk CTF) as representing the least complex system in which the binding of exogenous peptides could be studied. ΔStalk CTF is not a biological form or a mutant protein of PC1 observed in ADPKD. However, in our previous study, it mimicked the ADPKD-associated stalk mutants by being defective in cell signaling assays and rarely formed the R3848-E4078 salt bridge that was frequently seen in GaMD simulations with wild type CTF²⁸. Therefore ΔStalk CTF served as the negative control for the following studies.

The cryo-EM structure of human PC1-PC2 complex (PDB: 6A70)⁴⁷ was used to build the computational model for ΔStalk PC1 CTF after deleting the first 21 residues (3049-3069) from the CTF²⁸. We successfully docked the p9, p17 and p21 stalk peptides to the ΔStalk CTF model with HPEPDOCK⁴⁸ (**See SI**). The peptides all bound to the TOP domain and the interface between the TOP domain and extracellular loop 1 (ECL1) of CTF (**Fig. S3A-B**). In particular, peptide p21 occupied a closely similar binding region as the stalk in wildtype CTF as observed in the previous study²⁸. We then performed five independent 500 ns Pep-GaMD simulations on each of the three stalk peptide agonists p9, p17 and p21 bound to ΔStalk CTF to refine their HPEPDOCK docking conformations (**See SI**).

With the Pep-GaMD simulation frames, we performed structural clustering of each peptide using the hierarchical agglomerative algorithm in CPPTRAJ⁴⁹. The top-ranked conformations of each peptide bound to ΔStalk CTF were compared to their initial docking conformations. Next, we calculated 2D free energy profiles of the peptides-bound ΔStalk CTF by reweighting the Pep-GaMD simulations. The R3848-E4078 residue distance and the number of contacts between the peptides

(p9, p17 and p21) and TOP domains were selected as the reaction coordinates. The number of contacts was calculated between any atom pairs within 4 Å distance of the peptide and extracellular domains of PC1 protein. In the subsequent analyses, stalk and peptide residues are numbered relative to the N terminus of the stalk as starting from 1, while residues of the Δstalk CTF are numbered according to the human PC1 protein sequence.

Active conformation of peptide p9-bound PC1 CTF

From the free energy profile of the p9-bound ΔStalk CTF, we identified “Unbound” and “Bound” low-energy states (**Fig. 2A**). In the docking conformation, peptide p9 bound to the interface between the TOP and ECL1 of ΔStalk CTF (**Fig. S3A-B**). In Pep-GaMD simulations, the p9 peptide dissociated from the TOP-ECL1 binding pocket and rebound to the TOP domain in a slightly different region (**Fig. 2B-C**). The p9 peptide sequence is mostly composed of hydrophobic residues. Polar interactions between the main chain atoms of peptide-protein residues were observed in the top-ranked representative conformation of the p9-bound ΔStalk CTF. Protein residues R3892 and H3864 formed hydrogen bonds with p9 residues A2 and A5, respectively (**Fig. 2D**). These interactions were also highlighted in the protein contact map between peptide p9 and the extracellular domains of CTF in the representative “Bound” state (**Fig. S4**). The distance between the TOP domain residue R3848 and PL residue E4078 was 3.9 Å (**Fig. 2E**), suggesting that the top-ranked representative conformation of the p9-bound ΔStalk CTF was in the “Closed/Active” low-energy state. In addition, the 2D free energy profile of each individual simulation was calculated and the “Bound” low-energy state was consistently identified in the 2D free energy profiles of peptide p9 (**Fig. S5**).

Active and Intermediate conformational states of peptide p17-bound PC1 CTF

The free energy profile of the p17-bound ΔStalk system allowed us to identify three low-energy states - “Unbound”, “Intermediate”, and “Bound” (**Fig. 3A**). In the docking conformation, peptide p17 bound to the interface between the TOP and ECL1 of ΔStalk CTF (**Fig. S3A-B**). In the Pep-GaMD refined “Bound” state, a folded antiparallel β-strand conformation was observed for the peptide p17 at the interface of ECL1 and the TOP domain (**Fig. 3B-C**). Peptide residues T1, F3, A5, F8, F16 and V17 formed hydrophobic interactions with the protein residues H3311, R3314 and Y3307 from ECL1, and E3708, S3711, Q3707, A3704, R3700 and L3701 from the TOP domain (**Fig. 3D**). These interactions were also highlighted in the protein contact map between peptide p17 and the extracellular domains of CTF in the representative “Bound” state (**Fig. S4**). The distance between the TOP domain residue R3848 and PL residue E4078 was 4.1 Å (**Fig. 3E**), suggesting that the top-ranked representative conformation of the p17 bound ΔStalk CTF was in the “Closed/Active” low-energy state.

In the “Intermediate” state, p17 with a short helical turn was also observed to bind the TOP domain of ΔStalk CTF (**Fig. 3B-C**). Hydrophobic residue interactions were also formed between the peptide and protein. In particular, peptide residues T1, F3, P10, P11, H13, R15, F16 and V17 formed hydrophobic interactions with the protein residues P3859, A3704, S3741, Q3739, Y3734, P3733, H3729, W3726, R3712 and R3856 from the TOP domain (**Fig. 3F**). The distance between the TOP domain residue R3848 and PL residue E4078 was 14.6 Å (**Fig. 3G**), suggesting that this representative conformation (ranked the second among the Pep-GaMD structural clusters) of the p17-bound ΔStalk CTF was in the “Intermediate” low-energy state. In addition, the 2D free energy profile of each individual simulation was calculated. Pep-GaMD simulations were able to refine the peptide conformation from the “Unbound” to “Intermediate” and “Bound” states in Sim1 and Sim5, while the peptide reached only the “Intermediate” state in the other three simulations (**Fig. S6**). The free energy values of 2D PMF minima shown in **Fig. 3A** could differ from those in the 1D PMF minima of peptide structural clusters, especially with the usage of distinct reaction coordinates.

Figure 2.

(A) Free energy profile of the p9-bound Δ Stalk CTF regarding the number of atom contacts between p9 and extracellular domains of CTF and the distance between the CZ atom of R3848 and the CD atom of R4078 in CTF calculated from Pep-GaMD simulations. (B-C) Comparison of HPEPDOCK docking (cyan) and Pep-GaMD refined (magenta) conformations of peptide p9 in Δ Stalk CTF. (D) Polar interactions between peptide-protein residues observed in the top-ranked representative conformations of p9. Peptide residues are numbered relative to the N terminus of the stalk with the peptide starting from 1, while residues within Δ stalk CTF are numbered according to the standard PC1 residue number. (E) Distance between TOP domain residue R3848 and PL residue E4078 observed in p9-bound Δ Stalk CTF.

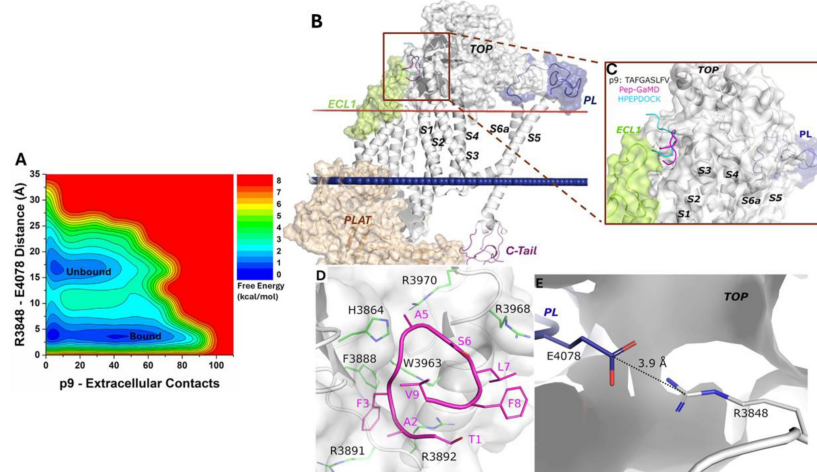
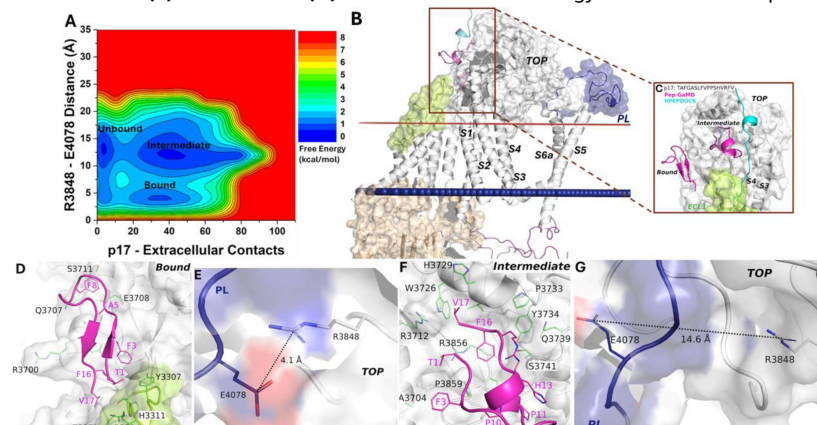


Figure 3.

(A) Free energy profile of the p17-bound Δ Stalk CTF regarding the number of atom contacts between p17 and extracellular domains of CTF and the distance between the CZ atom of R3848 and the CD atom of R4078 in CTF calculated from Pep-GaMD simulations. (B-C) Comparison of HPEPDOCK docking (cyan) and Pep-GaMD refined (magenta) conformations of peptide p17 in Δ Stalk CTF. Hydrophobic interactions (red dashed lines) between peptide-protein residues observed in the (D) "Bound" and (F) "Intermediate" low-energy conformations of p17-bound Δ Stalk CTF. Distance between TOP domain residue R3848 and PL residue E4078 observed in the (E) "Bound" and (G) "Intermediate" low-energy conformations of p17-bound Δ Stalk CTF.



Active conformational state of peptide p21-bound PC1 CTF

Finally, the free energy profile of the p21-bound Δ Stalk CTF allowed us to identify only a broad low-energy well corresponding to the “Bound” state (Fig. 4A). The docking conformation of p21-bound Δ Stalk CTF was refined through Pep-GaMD simulations, where folding of the peptide was observed on the protein surface of the TOP domain (Fig. 4B–C). The p21 peptide occupied a similar binding region as the stalk in wildtype CTF as observed in the previous study²⁸. Hydrophobic contacts were observed between peptide residues L7, F8, P10, S12, H13, V14, V17, P19, E20 and P21 and protein residues L3863, L3701, I3705, L3709, E3708, R3712, F3714, H3729, W3726, V3730, L3732, P3733, N3738, R3856 and S3741 (Fig. 4D). These interactions were also highlighted in the protein contact map between peptide p21 and the extracellular domains of CTF in the representative “Bound” state (Fig. S4). The distance calculated from the top-ranked structural cluster of the system between the TOP domain residue R3848 and PL residue E4078 was 3.8 Å, corresponding to the “Closed/Active” low-energy state (Fig. 4E). Furthermore, time courses of the radius of gyration (R_g) and root-mean square deviation (RMSD) of p21 relative to the starting HPEPDOCK conformation showed large conformational changes of the peptide during Pep-GaMD simulations (Fig. S7). In addition, the 2D free energy profile of each individual simulation showed that Pep-GaMD was able to refine the peptide docking conformation to the “Bound” state in all the five simulations (Fig. S8).

Peptide binding regions correlated with covarying residue pairs identified between the TOP domain and stalk TA

To provide an independent basis of evidence supporting the observation of “Bound” and “Intermediate” states of agonist peptide binding, we constructed a multiple sequence alignment (MSA) with an effective count of 1022 evolutionarily diverged PC1 homologs (illustrated in Fig. S9) from which we inferred a Potts statistical model (Fig. 5). Columns of the MSA with “direct” statistical interactions, as detected using the Potts inference method, reflect compensatory mutation pairs maintained through evolution supporting a conserved function. We limited our MSA to 394 residues on the extracellular side of PC1 because of the computational challenge of fitting the entire PC1 sequence (Fig. 5B and Fig. S9).

Fig. 5A shows the pairs of positions with strong Potts interaction scores where one position is either in the stalk, the nearby GAIN domain, or the TM1 helix. Some predicted interaction pairs recapitulated beta-sheet contacts within the GAIN domain observed in the homolog rat latrophilin-1¹¹ as well as predicted by AlphaFold⁵⁰ (Fig. S10) or between the extracellular ends of the TM2/4 and TM5/6 alpha helices known from cryo-EM structures⁴⁷ or predicted by AlphaFold, validating that our model detected biologically functional interactions.

We identified strong interactions between the stalk and other residues from the Potts model. They were not observed in the cryo-EM structure, in which the flexible stalk is missing. For interactions with the TOP domain, out of the 4875 possible pairs (25 stalk residues by 195 TOP domain residues in our Potts model), this analysis detected a stringent set of 6 strongly interacting pairs. Remarkably, multiple positions in this small set were among those relevant to the “Intermediate” binding conformation of p17 and “Bound” conformation of p21 as identified from the Pep-GaMD simulations. These were W3726 and S3741 in the TOP domain, both interacting with T1 of the stalk, and P3859 interacting with N25 at the end of the stalk (Fig. 5A). Additionally, we identified E3743 to be strongly interacting with F16 in the stalk, and it was also near the observed binding region in the TOP domain for the peptide p17 in the “Intermediate” state near S3741 (Fig. 3F). The remaining two strong interactions between the stalk and TOP domain involved Q3821 with T1 and L3893 with G4. L3893 is adjacent to R3892 that was identified to interact with the peptide p9 in

Figure 4.

(A) Free energy profile of the p21-bound Δ Stalk CTF regarding the number of atom contacts between p21 and extracellular domains of CTF and the distance between the CZ atom of R3848 and the CD atom of R4078 in CTF calculated from Pep-GaMD simulations. (B-C) Comparison of HPEPDOCK docking (cyan) and Pep-GaMD refined (magenta) conformations of peptide p21 in Δ Stalk CTF. (D) Polar interactions between peptide-protein residues observed in the top-ranked representative conformations of p21. (E) Distance between TOP domain residue R3848 and PL residue E4078 observed in p21-bound Δ Stalk CTF.

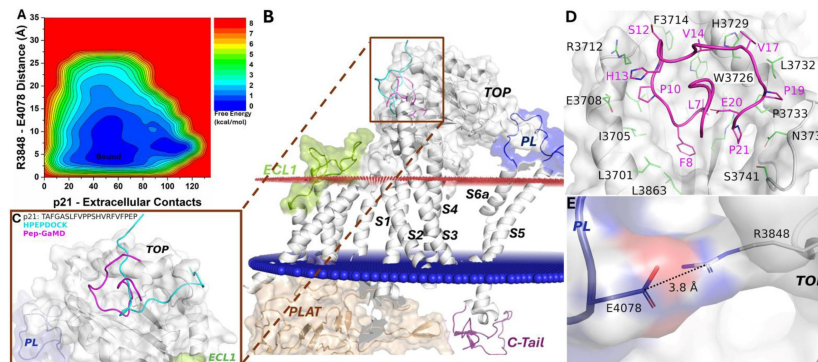
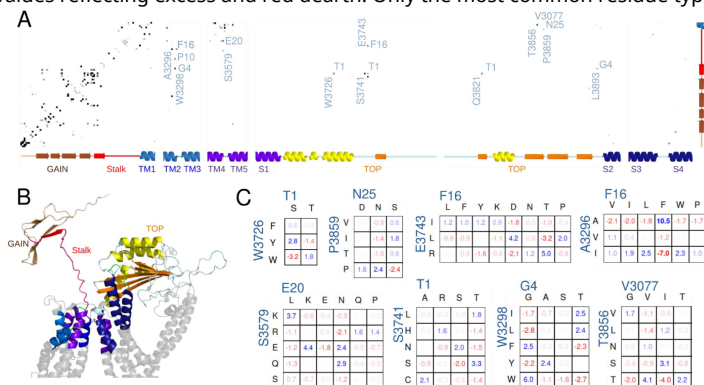


Figure 5.

(A) Potts interaction map based on the PKD1 multiple-sequence-alignment illustrated in **Figure S4**, showing interactions with the stalk. Gray dots are shown for residue position-pairs with Potts covariation scores above a threshold, colored darker for higher scores, and selected interacting pairs are annotated with the stalk residue (horizontal, numbered from the stalk N-terminus) and other residue (vertical, standard PKD1 numbering) with the PKD1 residue at each position. The secondary structure as a function of position is annotated along the axes. (B) Cartoon showing the subset of PC1 included in the Potts covariation analysis colored as in the secondary structure in panel A, using a structure predicted by AlphaFold. Gray regions were excluded from the Potts model. (C) Residue Covariation scores for selected position-pairs. The scores reflect the percentage excess frequency of the residue-pair relative to the null expected frequency if the MSA columns were uncorrelated, with blue values reflecting excess and red dearth. Only the most common residue types are shown.



the “Bound” state (**Fig. 2D**) and mutating it may affect its neighbor’s positioning. Besides the interactions between the TOP domain and the stalk TA, we also found a set of interactions between the stalk and the extracellular ends of TM2-TM3 helices and TM4-

TM5 helices, in which stalk residues G4, P10, F16 and E20 interact with W3298, A3296, and S3579, respectively, as well as a strong interaction between V3077, three residues past the end of the stalk, and T3856 in the TOP domain. TOP domain residue T3856 was also identified as relevant to the binding region of peptide p17 in the “Intermediate” state (**Fig. 3F**) and peptide p21 in the “Bound” state (**Fig. 4D**). These interactions could additionally play a role in stalk-TA activation or could be related to other functionality such as cleavage in the GPS motif.

To gain further insight and to validate that these detected “direct” interactions reflect biologically meaningful functional interactions and are not artifacts of the data, we examined the residue-specific covariation observed in the MSA (**Fig. 5C**), which measures the difference between the observed pairwise residue frequency and its null expectation under assumption of independent variation. Values greater than ~1% are commonly found to be indications of a statistically reliable mutational covariation (see **SI Computational Methods**), and many of the covarying pairs discovered between the stalk and TOP domain were significantly above this value. We validated that the covarying residue pairings were consistent with biophysical interaction. For example, for the position-pair 20-3579, here annotated such that the first index is the stalk residue numbered relative to the N terminus of the stalk and the second index is the TOP domain residue numbered according to the human PC1 protein sequence, there were excess residue-pair counts in the MSA consistent with opposite-charge or polar pairing such as K20-E3579, N20-Q3579, and others, and a dearth of repulsive like-charge pairs such as E20-E3579. Similarly, position-pair 1-3741 favored certain combinations of polar residues such as T1-S3741. Other position-pairs appeared consistent with hydrophobic packing interactions, such as F16-A3296, G4-W3298, and T1-W3726. A large residue F or W at position 3298 in the TOP domain was commonly paired with a G at stalk position 4, while a smaller I or L residue at position 3298 was more commonly paired with T at stalk position 4.

Molecular Mechanics/Poisson-Boltzmann Surface Area (MM/PBSA)⁵¹ analysis was further performed to calculate the binding free energies of peptides p9, p17 and p21 to PC1 CTF and decompose the residue-wise energy contributions using the gmx_MMPBSA software⁵². The relative rank of the overall peptide binding free energies (**Table S1**) was consistent with the experimental signaling data, i.e., p21>p9>p17, for which p21 showed the largest binding free energy value of binding (-40.29 ± 6.94 kcal/mol). Furthermore, we performed residue-wise energy decomposition analysis with MM/PBSA using gmx_MMPBSA software⁵², which allowed us to identify key residues that contributed the most to the peptide binding energies. These included residues T1 and V9 in p9 (**Table S2**), residues T1, R15 and V17 in p17 (**Table S3**), and residues P10, P11, P19 and P21 in p21 and residue W3726 in the PC1 CTF (**Table S4**). The energetic contributions of these residues apparently correlated to the sequence coevolution predicted from the Potts model.

Discussion

In *in vitro*, cell-based signaling assays, PC1 CTF-mediated activation of the NFAT reporter is dependent on its N-terminal, extracellular stalk, as shown by the loss of reporter activity with the CTF stalk-deletion expression construct, CTF^{Δst} 28 (**Fig. 1B**) and by the ability of synthetic, stalk sequence-derived peptides to reactivate signaling by CTF^{Δst} *in trans* (**Fig. 1E**). A series of synthetic peptides derived from the N-terminal sequence of the mouse PC1 CTF stalk were used to determine their agonistic activity in PC1 CTF^{Δst}-transfected cells. Notably, treatment with stalk peptides p7, p9, p17, p19 and p21 resulted in significant NFAT reporter activity over CTF^{Δst} control (no peptide treatment), wherein the effects of p7, p9 and p17 were specific to mouse CTF^{Δst}.

expressing cells. These data are consistent with the stalk peptides acting as soluble TA peptide agonists for PC1 and provide further evidence for the activation of PC1 signaling via an ADGR-like TA mechanism.

In ADPKD, numerous missense mutations reported within the GPCR autoproteolysis-inducing (GAIN) domain that have been shown to prevent or inhibit cleavage at the GPCR-coupled proteolysis site (GPS)¹¹. Loss of PC1 GPS cleavage, which is known to cause ADPKD, would likely sequester the stalk tethered agonist within the interior of the GAIN domain, which would presumably interfere with interactions between stalk tethered agonist residues and the remainder of the CTF and thus loss of the stalk-mediated signaling mechanism. Furthermore, there are 10 single nucleotide polymorphisms reported within the stalk sequence (ADPKD Variant Database; <https://pkdb.mayo.edu/welcome>), most of which were found to significantly reduce CTF-mediated activation of the NFAT reporter⁵³. In particular, the ADPKD-associated G3052R stalk mutation that was previously analyzed along with the stalkless CTF by GaMD simulations²⁸ has the same reduction in activity as the stalkless CTF in the cellular signaling reporter assays and the same loss of active/closed conformation interactions in GaMD analyses. Therefore, the stalkless CTF was used in our docking and simulation studies as representative of a biologically relevant mutant form of PC1.

To reveal the molecular mechanisms of the soluble stalk-derived peptides, we chose to perform HPEPDOCK docking and novel Pep-GaMD simulations to sample the peptide interactions with the Δ Stalk PC1 CTF. Pep-GaMD simulations were able to refine the docking conformations of peptide agonists bound to the Δ Stalk PC1 CTF. It is important to note that the free energy profiles calculated from GaMD simulations of PC1 CTF were not fully converged since certain variations were observed among the individual simulations. Nevertheless, these calculations allowed us to identify representative low-energy binding conformations of the peptides. Pep-GaMD simulations sampled an antiparallel β -strand and a short helical secondary structure of peptide p17 bound to the Δ Stalk CTF. Furthermore, peptides p9 and p21 adopted a more folded structure as compared to their disordered loop conformations in the docking poses. We also observed TOP-PL interactions, particularly the salt bridge between residues R3848-E4078 that is a key feature of the stalk TA-mediated activation of signaling for PC1 CTF²⁸. Signal transduction was initiated upon binding of the stalk (TA) to the TOP domain, which was transmitted to the PL via a salt bridge formation between residue R3848 in the TOP domain and residue E4078 in the PL. The bound peptide agonists p9, p17 and p21 maintained the Δ Stalk CTF in its “Closed/Active” conformation as observed in the wildtype PC1 CTF simulations²⁸.

The interacting pairs identified using sequence-based covariation analysis matched the pairs identified by Pep-GaMD simulations, providing complementary evidence of the importance of these interactions and of the existence of the “Bound” and “Intermediate” binding states of the stalk TA and stalk-derived peptide agonist. This suggests that such stalk TA binding states are evolutionarily conserved across PC1 orthologs. Covariation analysis identifies interactions important during any part of the protein lifecycle, and alone cannot be used to distinguish which conformational state an interaction arises in. By comparison to the conformations found in the Pep-GaMD simulations, we found that most of the identified interactions between the stalk TA and TOP domain were consistent with either the “Intermediate” or “Bound” binding states of the stalk-derived peptides, which are related to CTF inactive and active signaling states, however, it remained possible that other interactions, such as between the start of the stalk-TA and TM2/TM3, may be related to conformational states necessary for cleavage of the GAIN/GPS domain. Additionally, structural contacts may be incompletely detected at some positions when the statistical signal of covariation is masked by high conservation, subfamily specialization, or misalignment. This can explain why some interactions identified in the binding interface through docking are not detected using covariation analysis. Despite this, the specific subset of interactions detected using covariation analysis suggest broader peptide binding interfaces, and we found

these to be consistent with the peptide binding interactions observed in the Pep-GaMD simulations and the MM/PBSA binding free energy analysis, and the covarying residue pairings were consistent with functional biophysical interactions.

The proposed binding interactions of the PC1 stalk peptides shares some similarity with those observed for the ADGRs. Specifically, Xiao et al. resolved cryo-EM structures of active ADGRG2 and ADGRG4 in complex with tethered Stachel sequences²⁷. The structures showed that the 15 residue Stachel sequence inserts into the TM bundle to form intense hydrophobic interactions. A hydrophobic F/Y/LX $\phi\phi\phi$ X ϕ motif identified in the ADGR tethered sequences formed five finger-like projections in the hydrophobic pits of the TM bundle²⁷. In our study, we observed a similar pattern of intense hydrophobic interactions between the peptide agonists p9, p17 and p21 and the hydrophobic pockets in the TOP domain of PC1 CTF. Notably, a closely similar TOP binding pocket was identified for interaction of the tethered agonist (Stalk) in our previous study²⁸ and for binding of peptide agonist p21 in this study. The TOP domain hydrophobic pocket may serve as a significant candidate binding site for designing new synthetic peptides or small molecules to aid in rescue of PC1 function levels. Moreover, the shorter peptide agonists' (p9 and p17) binding sites also serve as novel pockets for design and development of therapeutic approaches for treating ADPKD. While the present study is focused on identification of initial peptides that are able to activate the PC1 CTF, we shall include further mutation experiments and simulations, peptide SAR and optimization of the lead peptides in future studies. It is also important to note that we have not tested the selectivity of the peptides for PC1 versus PC2 in the present study primarily because transfection of PC2 does not activate the NFAT reporter. However, it is possible that co-transfection of PC2 with the PC1 CTF could alter the stalk peptide binding. This will be important to consider in future studies.

Acknowledgements

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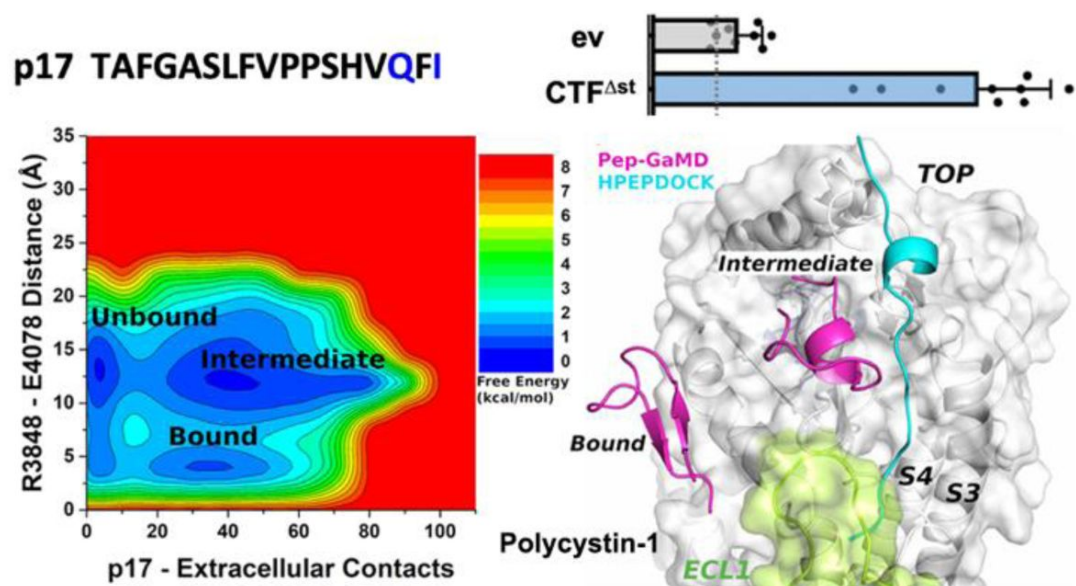
Author Contributions

R.M. and Y.M. designed research; S.P., B. S. M., E. N. M. and A.H. performed research; S.P., B. S. M., K.J.; E. N. M., A.H., R.M. and Y.M. analyzed data; and S.P., K.J.; A.H., R.M. and Y.M. wrote the paper.

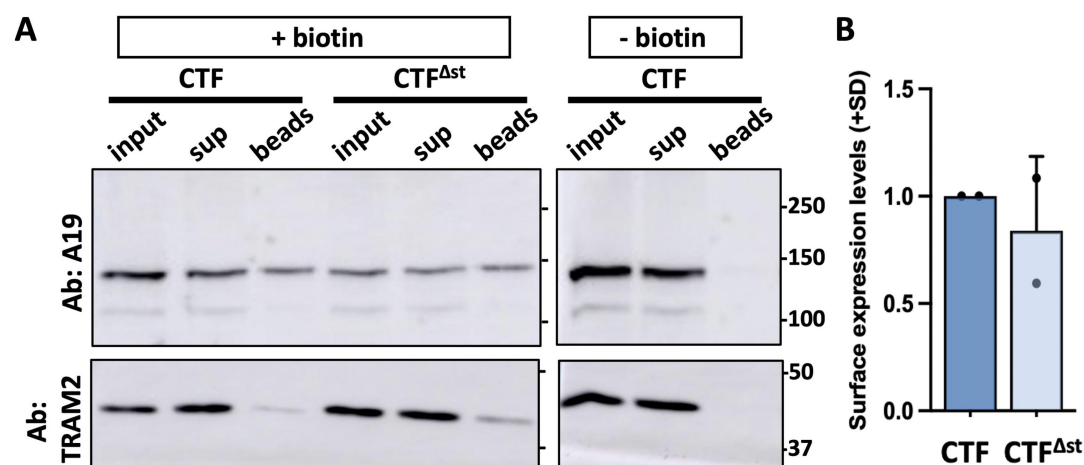
Competing Interest Statement

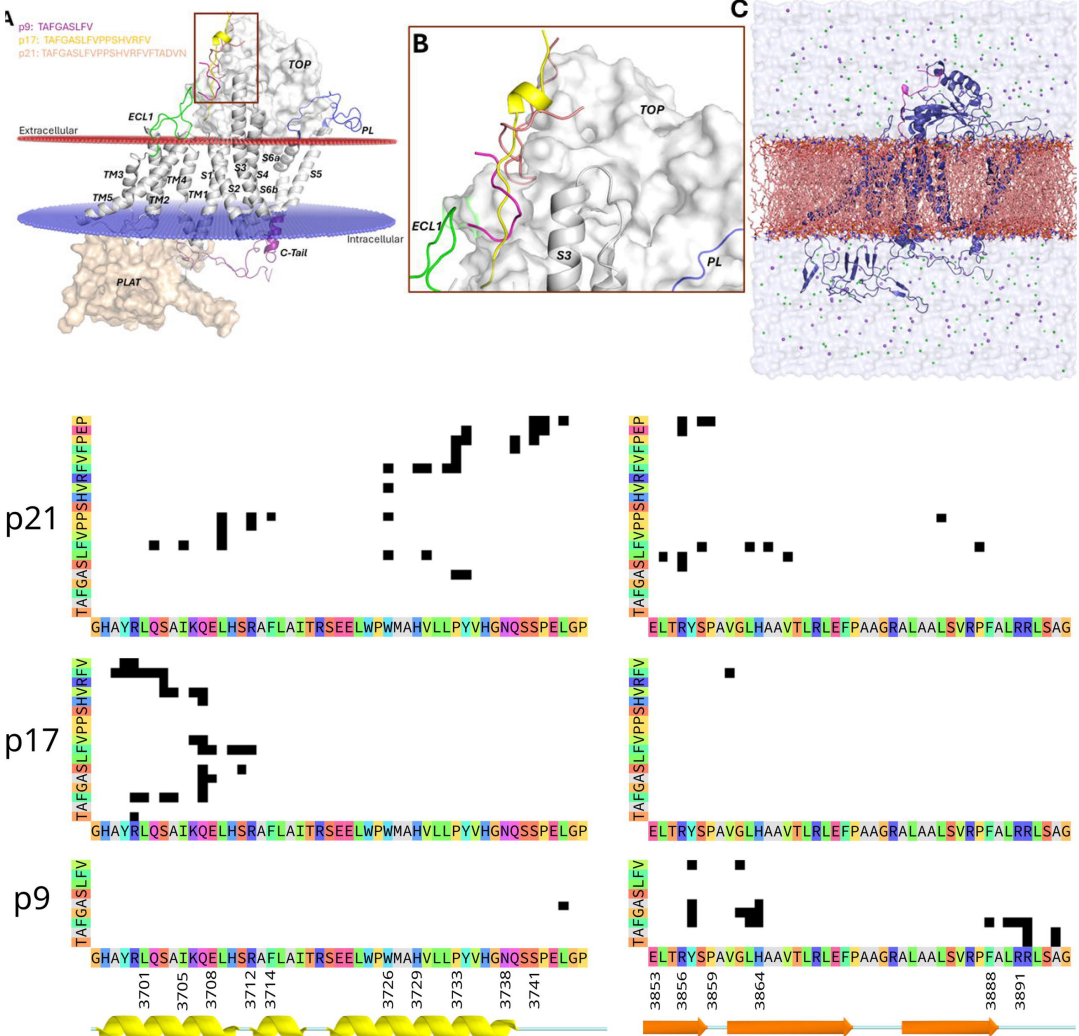
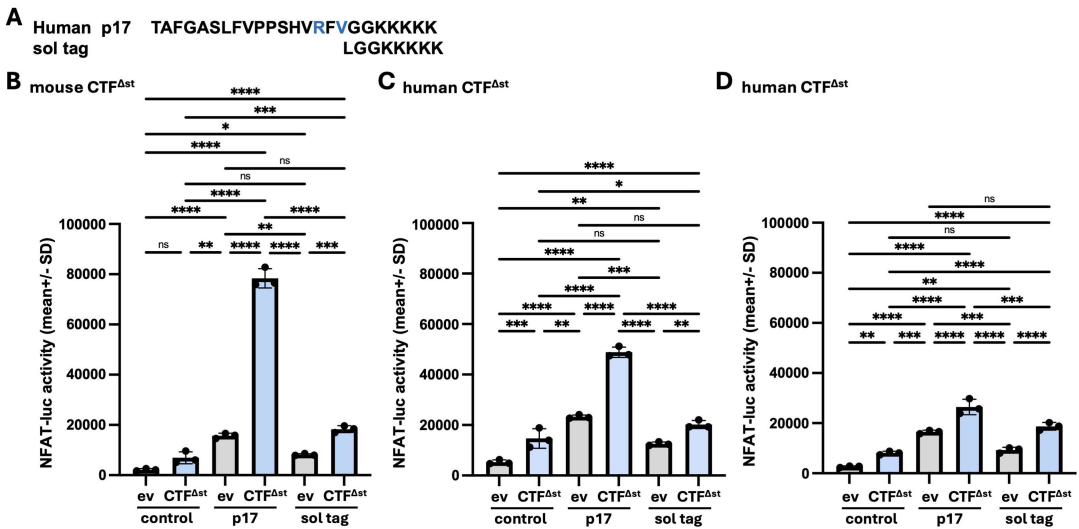
No competing interests.

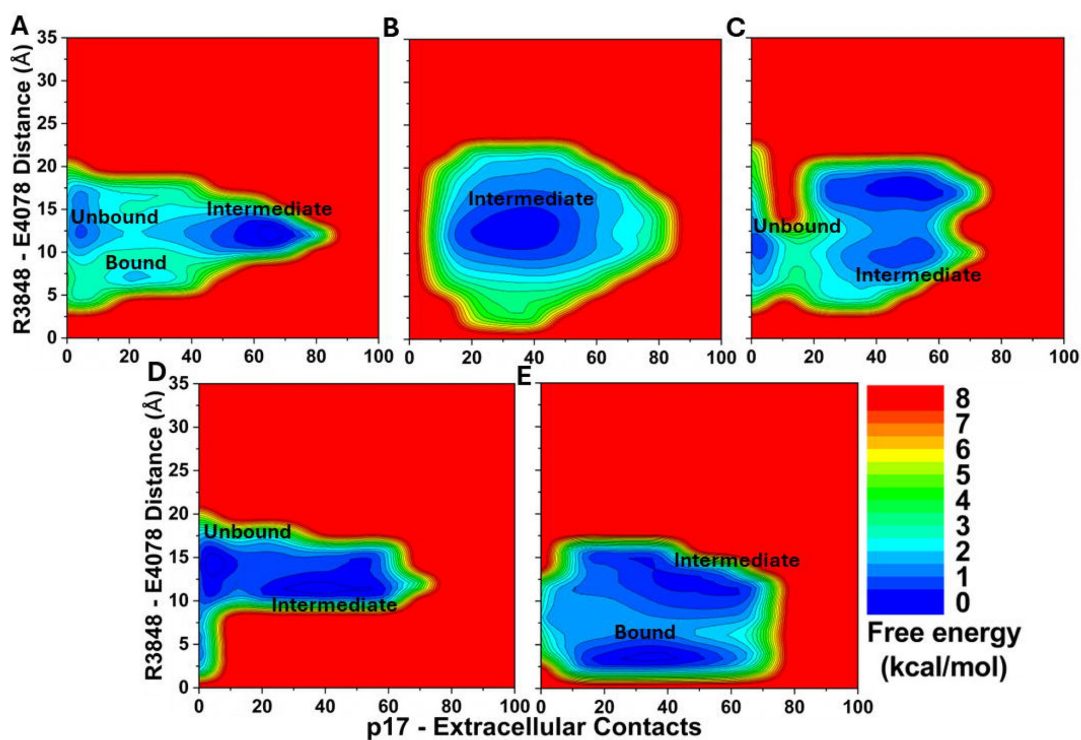
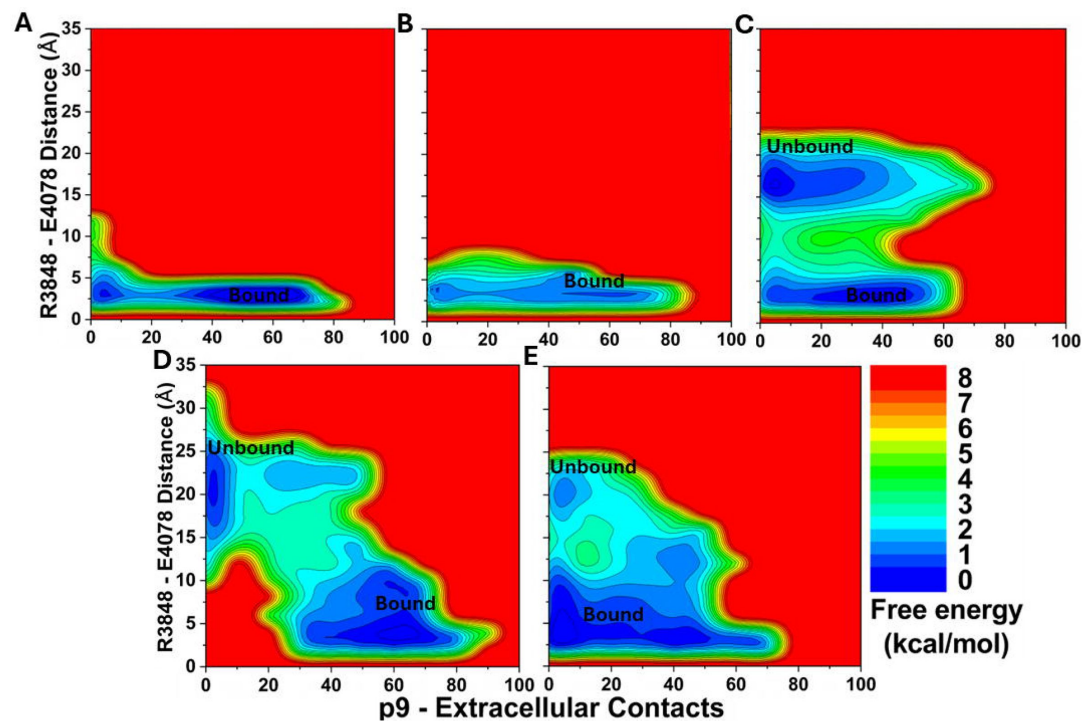
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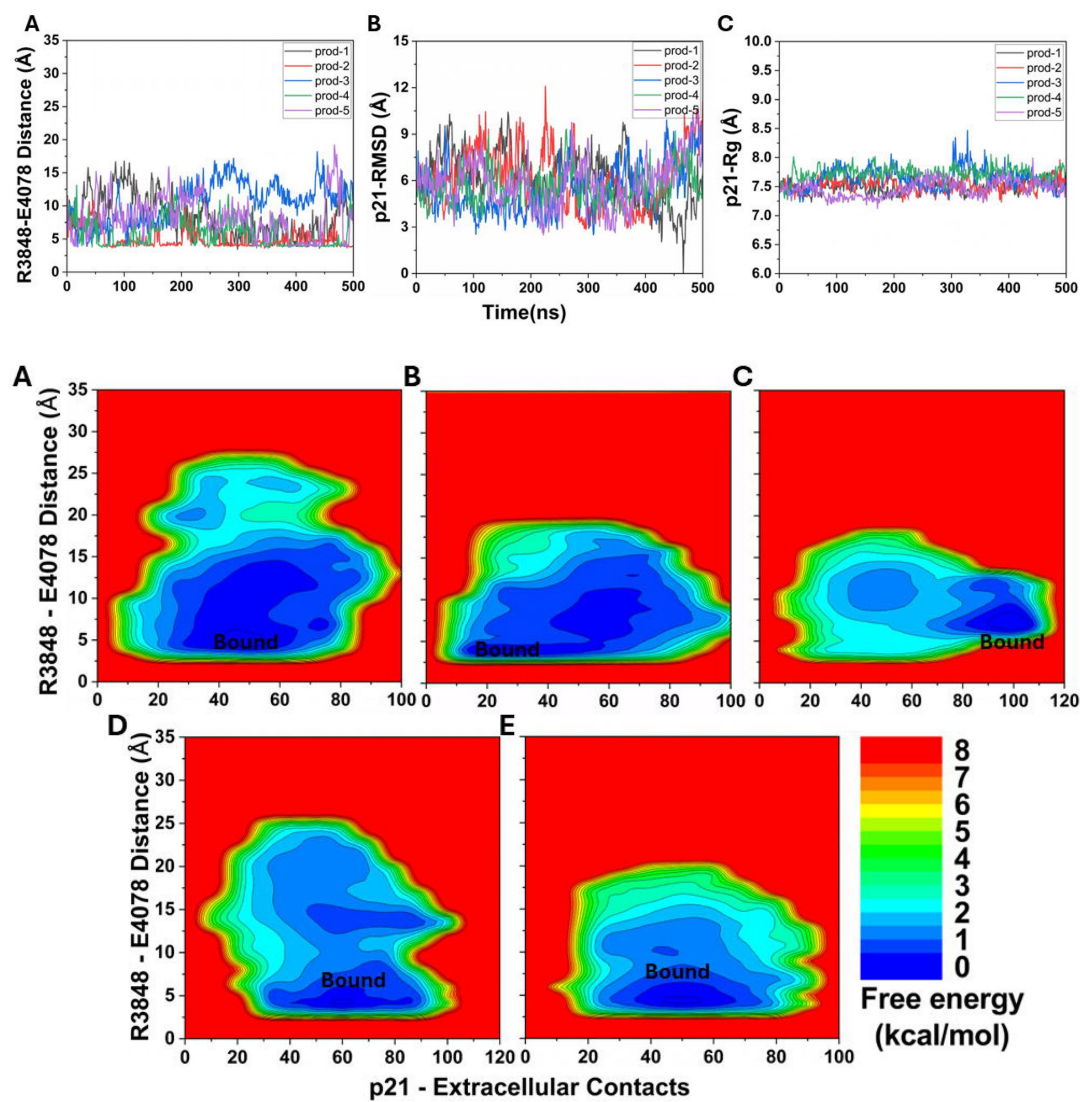


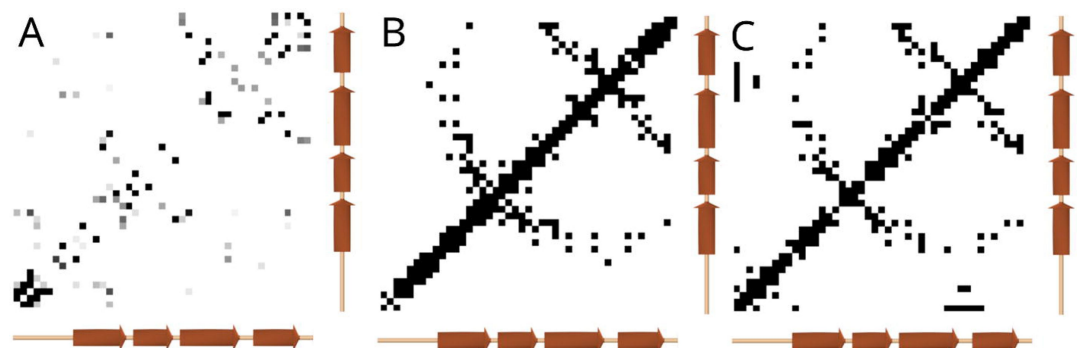
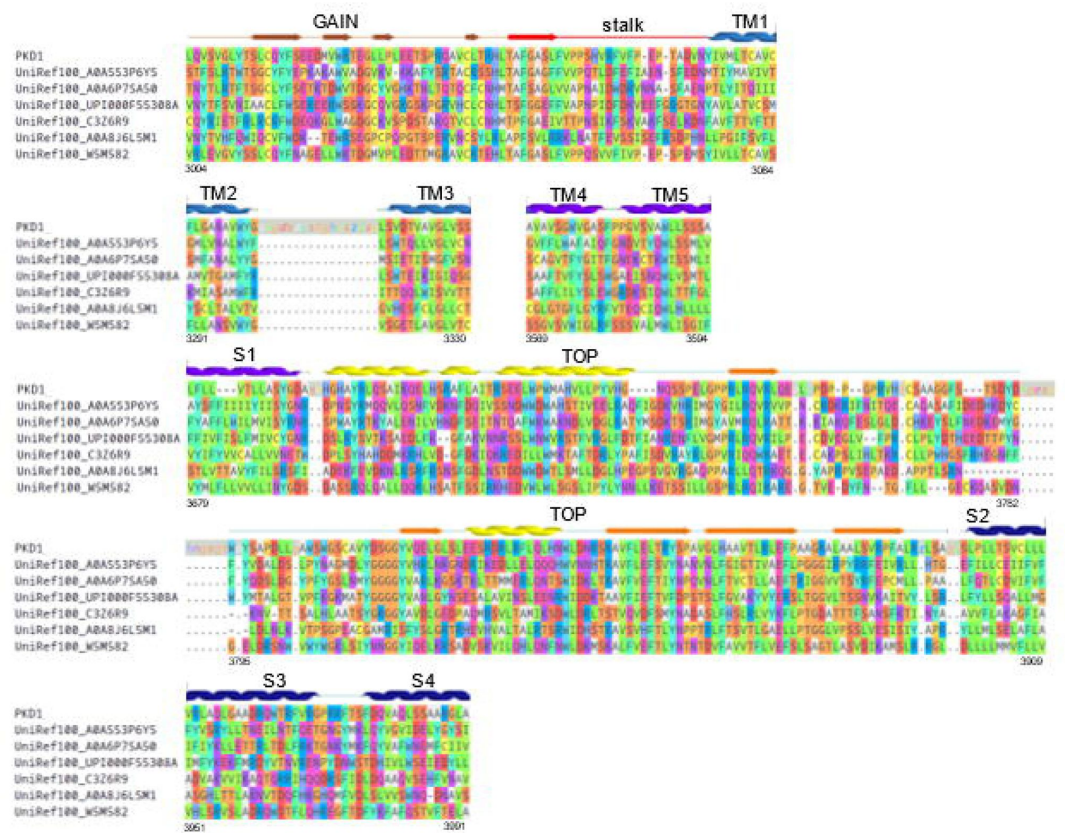
Structural dynamic models are presented for binding of novel synthetic, soluble peptide agonists and associated activation of Polycystin-1 through a combination of complementary cellular signaling assays, accelerated molecular simulations and sequence coevolutionary analysis. The p17 peptide derived from the first 17 residues of the protein stalk tethered agonist is shown here.











References

- (1) Harris P. C., Torres V. E (2014) **Genetic mechanisms and signaling pathways in autosomal dominant polycystic kidney disease** *The Journal of clinical investigation* **124**:2315–2324
- (2) Ingelfinger J. R (2017) **Tolvaptan and autosomal dominant polycystic kidney disease** *Mass Medical Soc* **377**:1988–1989
- (3) Hopp K., Ward C. J., Hommerding C. J., Nasr S. H., Tuan H.-F., Gainullin V. G., Rossetti S., Torres V. E., Harris P. C (2012) **Functional polycystin-1 dosage governs autosomal dominant polycystic kidney disease severity** *The Journal of clinical investigation* **122**:4257–4273
- (4) Cai Y., Fedeles S. V., Dong K., Anyatonwu G., Onoe T., Mitobe M., Gao J.-D., Okuhara D., Tian X., Gallagher A.-R (2014) **Altered trafficking and stability of polycystins underlie polycystic kidney disease** *The Journal of clinical investigation* **124**:5129–5144
- (5) Hofherr A., Wagner C. J., Watnick T., Köttgen M (2016) **Targeted rescue of a polycystic kidney disease mutation by lysosomal inhibition** *Kidney international* **89**:949–955
- (6) Krappitz M., Gallagher A.-R., Fedeles S (2016) **Is it time to fold the cysts away?** *Trends in molecular medicine* **22**:997–999
- (7) Lakhia R., Ramalingam H., Chang C.-M., Cobo-Stark P., Biggers L., Flaten A., Alvarez J., Valencia T., Wallace D. P., Lee E. C (2022) **PKD1 and PKD2 mRNA cis-inhibition drives polycystic kidney disease progression** *Nature communications* **13**
- (8) Tan Y.-C., Blumenfeld J., Rennert H (2011) **Autosomal dominant polycystic kidney disease: genetics, mutations and microRNAs** *Biochimica et Biophysica Acta (BBA)-Molecular Basis of Disease* **1812**:1202–1212
- (9) Rossetti S., Kubly V. J., Consugar M. B., Hopp K., Roy S., Horsley S. W., Chauveau D., Rees L., Barratt T. M., Van't Hoff W. G (2009) **Incompletely penetrant PKD1 alleles suggest a role for gene dosage in cyst initiation in polycystic kidney disease** *Kidney international* **75**:848–855
- (10) Heyer C. M., Sundsbak J. L., Abebe K. Z., Chapman A. B., Torres V. E., Grantham J. J., Bae K. T., Schrier R. W., Perrone R. D., Braun W. E (2016) **Predicted mutation strength of nontruncating PKD1 mutations aids genotype-phenotype correlations in autosomal dominant polycystic kidney disease** *Journal of the American Society of Nephrology* **27**:2872–2884
- (11) Araç D., Boucard A. A., Bolliger M. F., Nguyen J., Soltis S. M., Südhof T. C., Brunger A. T. (2012) **A novel evolutionarily conserved domain of cell-adhesion GPCRs mediates autophosphorylation** *The EMBO journal* **31**:1364–1378
- (12) Qian F., Boletta A., Bhunia A. K., Xu H., Liu L., Ahrabi A. K., Watnick T. J., Zhou F., Germino G. G (2002) **Cleavage of polycystin-1 requires the receptor for egg jelly domain and is disrupted by human autosomal-dominant polycystic kidney disease 1-associated mutations** *Proceedings of the National Academy of Sciences* **99**:16981–16986

- (13) Kurbegovic A., Kim H., Xu H., Yu S., Cruanès J., Maser R. L., Boletta A., Trudel M., Qian F (2014) **Novel functional complexity of polycystin-1 by GPS cleavage in vivo: role in polycystic kidney disease** *Molecular and cellular biology* **34**:3341–3353
- (14) Kim S., Nie H., Nesin V., Tran U., Outeda P., Bai C.-X., Keeling J., Maskey D., Watnick T., Wessely O (2016) **The polycystin complex mediates Wnt/Ca²⁺ signalling** *Nature cell biology* **18**:752–764
- (15) Weston B. S., Bagnérès C., Price R. G., Stirling J. L (2001) **The polycystin-1 C-type lectin domain binds carbohydrate in a calcium-dependent manner, and interacts with extracellular matrix proteins in vitro** *Biochimica Et Biophysica Acta (BBA)-Molecular Basis of Disease* **1536**:161–176
- (16) Sandford R., Sgotto B., Aparicio S., Brenner S., Vaudin M., Wilson R. K., Chissoe S., Pepin K., Bateman A., Chothia C (1997) **Comparative analysis of the polycystic kidney disease 1 (PKD1) gene reveals an integral membrane glycoprotein with multiple evolutionary conserved domains** *Human molecular genetics* **6**:1483–1489
- (17) Ibraghimov-Beskrovnaya O., Bukanov N. O., Donohue L. C., Dackowski W. R., Klinger K. W., Landes G. M (2000) **Strong homophilic interactions of the Ig-like domains of polycystin-1, the protein product of an autosomal dominant polycystic kidney disease gene, PKD1** *Human molecular genetics* **9**:1641–1649
- (18) Weston B. S., Malhas A. N., Price R. G (2003) **Structure-function relationships of the extracellular domain of the autosomal dominant polycystic kidney disease-associated protein, polycystin-1** *FEBS letters* **538**:8–13
- (19) Nims N., Vassmer D., Maser R. L (2003) **Transmembrane domain analysis of polycystin-1, the product of the polycystic kidney disease-1 (PKD1) gene: evidence for 11 membrane-spanning domains** *Biochemistry* **42**:13035–13048
- (20) Maser R. L., Calvet J. P (2020) **Adhesion GPCRs as a paradigm for understanding polycystin-1 G protein regulation** *Cellular signalling* **72**
- (21) Maser R. L., Calvet J. P., Parnell S. C (2022) **The GPCR properties of polycystin-1-A new paradigm** *Frontiers in Molecular Biosciences* **9**
- (22) Yu S., Hackmann K., Gao J., He X., Piontek K., García-González M. A., Menezes L. F., Xu H., Germino G. G., Zuo J (2007) **Essential role of cleavage of Polycystin-1 at G protein-coupled receptor proteolytic site for kidney tubular structure** *Proceedings of the National Academy of Sciences* **104**:18688–18693
- (23) Liebscher I., Schön J., Petersen S. C., Fischer L., Auerbach N., Demberg L. M., Mogha A., Cöster M., Simon K.-U., Rothmund S (2014) **A tethered agonist within the ectodomain activates the adhesion G protein-coupled receptors GPR126 and GPR133** *Cell reports* **9**:2018–2026
- (24) Schöneberg T., Liebscher I., Luo R., Monk K. R., Piao X (2015) **Tethered agonists: a new mechanism underlying adhesion G protein-coupled receptor activation** *Journal of Receptors and Signal Transduction* **35**:220–223
- (25) Demberg L. M., Rothmund S., Schöneberg T., Liebscher I (2015) **Identification of the tethered peptide agonist of the adhesion G protein-coupled receptor GPR64/ADGRG2** *Biochemical and biophysical research communications* **464**:743–747

- (26) Stoveken H. M., Hajduczuk A. G., Xu L., Tall G. G (2015) **Adhesion G protein-coupled receptors are activated by exposure of a cryptic tethered agonist** *Proceedings of the National Academy of Sciences* **112**:6194–6199
- (27) Xiao P., Guo S., Wen X., He Q.-T., Lin H., Huang S.-M., Gou L., Zhang C., Yang Z., Zhong Y.-N (2022) **Tethered peptide activation mechanism of the adhesion GPCRs ADGRG2 and ADGRG4** *Nature* **604**:771–778
- (28) Pawnikar S., Magenheimer B. S., Munoz E. N., Maser R. L., Miao Y (2022) **Mechanism of tethered agonist-mediated signaling by polycystin-1** *Proceedings of the National Academy of Sciences* **119**
- (29) Miao Y., Sinko W., Pierce L., Bucher D., Walker R. C., McCammon J. A (2014) **Improved reweighting of accelerated molecular dynamics simulations for free energy calculation** *Journal of chemical theory and computation* **10**:2677–2689
- (30) Miao Y., Feher V. A., McCammon J. A (2015) **Gaussian accelerated molecular dynamics: unconstrained enhanced sampling and free energy calculation** *Journal of chemical theory and computation* **11**:3584–3595
- (31) Miao Y., McCammon J. A (2016) **Graded activation and free energy landscapes of a muscarinic G-protein-coupled receptor** *Proceedings of the National Academy of Sciences* **113**:12162–12167
- (32) Pang Y. T., Miao Y., Wang Y., McCammon J. A (2017) **Gaussian accelerated molecular dynamics in NAMD** *Journal of chemical theory and computation* **13**:9–19
- (33) Miao Y., McCammon J. A. (2017) **Gaussian accelerated molecular dynamics: theory, implementation, and applications** *Annual reports in computational chemistry* **13**:231–278
- (34) Wang Y.-T., Chan Y.-H (2017) **Understanding the molecular basis of agonist/antagonist mechanism of human mu opioid receptor through gaussian accelerated molecular dynamics method** *Scientific reports* **7**:1–11
- (35) Liao J.-M., Wang Y.-T (2019) **In silico studies of conformational dynamics of Mu opioid receptor performed using gaussian accelerated molecular dynamics** *Journal of Biomolecular Structure and Dynamics* **37**:166–177
- (36) Miao Y., Bhattarai A., Nguyen A. T., Christopoulos A., May L. T (2018) **Structural basis for binding of allosteric drug leads in the adenosine A1 receptor** *Scientific reports* **8**
- (37) Chuang C.-H., Chiou S.-J., Cheng T.-L., Wang Y.-T. (2018) **A molecular dynamics simulation study decodes the Zika virus NS5 methyltransferase bound to SAH and RNA analogue** *Scientific reports* **8**
- (38) Sibener L. V., Fernandes R. A., Kolawole E. M., Carbone C. B., Liu F., McAfee D., Birnbaum M. E., Yang X., Su L. F., Yu W (2018) **Isolation and visualization of a structural trigger that uncouples TCR signaling from pMHC binding** *Cell* **174**
- (39) Park J. B., Kim Y. H., Yoo Y., Kim J., Jun S.-H., Cho J. W., El Qaidi S., Walpole S., Monaco S., García-García A. A (2018) **Structural basis for arginine glycosylation of host substrates by bacterial effector proteins** *Nature communications* **9**

- (40) Miao Y., McCammon J. A (2018) **Mechanism of the G-protein mimetic nanobody binding to a muscarinic G-protein-coupled receptor** *Proceedings of the National Academy of Sciences* **115**:3036–3041
- (41) Ricci C. G., Chen J. S., Miao Y., Jinek M., Doudna J. A., McCammon J. A., Palermo G (2019) **Deciphering off-target effects in CRISPR-Cas9 through accelerated molecular dynamics** *ACS central science* **5**:651–662
- (42) Palermo G., Miao Y., Walker R. C., Jinek M., McCammon J. A (2017) **CRISPR-Cas9 conformational activation as elucidated from enhanced molecular simulations** *Proceedings of the National Academy of Sciences* **114**:7260–7265
- (43) Bhattarai A., Wang J., Miao Y (2020) **G-protein-coupled receptor–membrane interactions depend on the receptor activation state** *Journal of computational chemistry* **41**:460–471
- (44) Pawnikar S., Miao Y (2020) **Pathway and mechanism of drug binding to chemokine receptors revealed by accelerated molecular simulations** *Future Medicinal Chemistry* **12**:1213–1225
- (45) Wang J., Miao Y (2020) **Peptide Gaussian accelerated molecular dynamics (Pep-GaMD): Enhanced sampling and free energy and kinetics calculations of peptide binding** *The Journal of chemical physics* **153**
- (46) Levy R. M., Haldane A., Flynn W. F (2017) **Potts Hamiltonian models of protein co-variation, free energy landscapes, and evolutionary fitness** *Current opinion in structural biology* **43**:55–62
- (47) Su Q., Hu F., Ge X., Lei J., Yu S., Wang T., Zhou Q., Mei C., Shi Y (2018) **Structure of the human PKD1-PKD2 complex** *Science* **361**
- (48) Zhou P., Jin B., Li H., Huang S.-Y (2018) **HPEPDOCK: a web server for blind peptide–protein docking based on a hierarchical algorithm** *Nucleic acids research* **46**:W443–W450
- (49) Roe D. R., Cheatham T. E (2013) **PTRAJ and CPPTRAJ: software for processing and analysis of molecular dynamics trajectory data** *Journal of chemical theory and computation* **9**:3084–3095
- (50) Jumper J., Evans R., Pritzel A., Green T., Figurnov M., Ronneberger O., Tunyasuvunakool K., Bates R., Žídek A., Potapenko A. (2021) **Highly accurate protein structure prediction with AlphaFold** *nature* **596**:583–589
- (51) Rastelli G., Rio A. D., Degliesposti G., Sgobba M (2010) **Fast and accurate predictions of binding free energies using MM-PBSA and MM-GBSA** *Journal of computational chemistry* **31**:797–810
- (52) Valdés-Tresanco M. S., Valdés-Tresanco M. E., Valiente P. A., Moreno E (2021) **gmx_MMPBSA: a new tool to perform end-state free energy calculations with GROMACS** *Journal of chemical theory and computation* **17**:6281–6291
- (53) Magenheimer B. S., Munoz E. N., Ravichandran J., Maser R. L (2021) **Constitutive signaling by the C-terminal fragment of polycystin-1 is mediated by a tethered peptide agonist** *bioRxiv*

Editors

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Senior Editor

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Reviewer #1 (Public Review):

Summary:

This research used cell-based signaling assay and Gaussian-accelerated molecular dynamics (GaMD) to study peptide-mediated signaling activation of Polycystin-1 (PC1), which is responsible for the majority of autosomal dominant polycystic kidney disease (ADPKD) cases. Synthetic peptides of various lengths derived from the N-terminal portion of the PC1 C-terminal fragment (CTF) were applied to HEK293T cells transfected with stalkless mouse CTF expression construct. It was shown that peptides including the first 7, 9, and 17 residues of the N-terminal portion could activate signaling to the NFAT reporter. To further understand the underlying mechanism, docking and peptide-GaMD simulations of peptides composed of the first 9, 17, and 21 residues from the N-terminal portion of the human PC1 CTF were performed. These simulations revealed the correlation between peptide-CTF binding and PC1 CTF activation characterized by the close contact (salt bridge interaction) between residues R3848 and E4078. Finally, a Potts statistical model was inferred from diverged PC1 homologs to identify strong/conserved interacting pairs within PC1 CTF, some of which are highly relevant to the findings from the peptide GaMD simulations. The peptide binding pockets identified in the GaMD simulations may serve as novel targets for design of therapeutic approaches for treating ADPKD.

Strengths:

- (1) The experimental and computational parts of this study complement and mostly support each other, thus increasing the overall confidence in the claims made by the authors.
- (2) The use of exogenous peptides and a stalkless CTF in the GaMD is a step forward compared to earlier simulations using the full CTF, CTF mutants, or the stalkless CTF alone. And it led to findings of novel binding pockets.
- (3) Since the PC1 shares characteristics with the Adhesion class of GPCRs, the approaches used in this work may be extended to other similar systems.

Weaknesses:

- (1) Only results for selective peptides (p9, p17 p21) binding with the protein were shown. It would be interesting to see the interaction between some (if not all) of the other peptides with the protein.
- (2) The convergence of the simulations is not very good. The results should be interpreted more qualitatively rather than quantitatively because large variations in the free energy profile were seen between different replicates. Although these simulations might have identified representative low-energy binding conformations of the peptides, whether they have explored all possible conformations is still a question.

<https://doi.org/10.7554/eLife.95992.2.sa3>

Reviewer #2 (Public Review):

Summary:

This manuscript, "Activation of Polycystin-1 Signaling by Binding of Stalk-derived Peptide Agonists", by Miao and coworkers. The autosomal dominant polycystic kidney disease (ADPKD) is a major form of polycystic kidney disease (PKD). To provide better treatment and avoid side effects associated with currently available options, the authors investigated an interesting GPCR, polycystin-1 (PC1), as a potential therapeutic target. In vitro and in silico studies were combined to identify peptide agonists for PC1 and to elucidate their roles in PC1 signaling. Overall, regarding the significance of the findings, this work described valuable peptide agonists for PC1 and the combined in vitro and in silico approach can be useful to study a complex system like PC1. However, the strength of the evidence is incomplete, as more experiments are needed as controls to validate the computational observations. The work appears premature.

Strengths:

- (1) This work first described the experimental discovery of short peptides designed to mimic the stalk region of PC1, followed by computational investigation using docking and MD simulations. PC1 is a complex membrane protein and an emerging target for ADPKD, but it can be challenging to study. The knowledge and the peptide discovery can be valuable and useful to understand the mechanism and potential modulation of PC1.
- (2) The authors published the mechanistic study of PC1 and identified key interacting residues such as N3074-S3585 and R3848-E4078, using very similar techniques (PNAS 2022, 119(19), e2113786119). This work furthers this research by identifying peptides that are stalk mimics for PC1 activation.
- (3) Eight peptides were designed and tested experimentally first; three were computationally studied with docking and GaMD simulations to understand their mechanism (s).

Weaknesses:

- (1) The selectivity of the peptides between PC1 and PC2 remains unknown in this revision.

Overall, my comments were mostly addressed properly.

<https://doi.org/10.7554/eLife.95992.2.sa2>

Reviewer #3 (Public Review):

Summary:

The authors demonstrate the activation of Polycystin-1 (PC1), a G-protein coupled receptor, using small peptides derived from its original agonist, the stalk TA protein. In the experimental part of the study, the authors performed cellular assays to check the peptide-induced reactivation of a mutant form of PC1 which does not contain the stalk agonist. The experimental data is supported by computational studies using state-of-the-art Gaussian accelerated Molecular Dynamics (GaMD) and bioinformatics analysis based on sequence covariance. The computer simulations revealed the mechanistic details of the binding of the said peptides with the mutant PC1 protein and discovered different bound, unbound, and intermediate conformations depending on the peptide size and sequence. Due to the use of reliable and well-established molecular simulation algorithms and the physiological relevance of this protein autosomal dominant polycystic kidney disease (ADPKD) make this work particularly valuable.

Strengths:

This work is exploratory and its goal is to establish that small peptides can be used to probe the PC1 signaling process. The authors have provided sufficient evidence to justify this claim. Their GaMD simulations have produced free-energy landscapes that differentiate the interaction of PC1 with three different synthetic peptides and demonstrate the associated conformational dynamics of the receptor protein. Their trajectory analysis and sequence covariance analysis could identify residue-specific interactions that facilitate this process. The authors also performed residue-wise and total interaction energy calculations to substantiate their findings.

Weaknesses:

The reported free energy landscapes are not fully converged. But they are still sufficient to gain biological insight.

<https://doi.org/10.7554/eLife.95992.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer 1:

This research used cell-based signaling assay and Gaussian-accelerated molecular dynamics (GaMD) to study peptide-mediated signaling activation of Polycystin-1 (PC1), which is responsible for the majority of autosomal dominant polycystic kidney disease (ADPKD) cases. Synthetic peptides of various lengths derived from the N-terminal portion of the PC1 C-terminal fragment (CTF) were applied to HEK293T cells transfected with stalkless mouse CTF expression construct. It was shown that peptides including the first 7, 9, and 17 residues of the N-terminal portion could activate signaling to the NFAT reporter. To further understand the underlying mechanism, docking and peptide-GaMD simulations of peptides composed of the first 9, 17, and 21 residues from the N-terminal portion of the human PC1 CTF were performed. These simulations revealed the correlation between peptide-CTF binding and PC1 CTF activation characterized by the close contact (salt bridge interaction) between residues R3848 and E4078. Finally, a Potts statistical model was inferred from diverged PC1 homologs to identify strong/conserved interacting pairs within PC1 CTF, some of which are highly relevant to the findings from the peptide GaMD simulations. The peptide binding pockets identified in the GaMD simulations may serve as novel targets for the design of therapeutic approaches for treating ADPKD.

We greatly appreciate the reviewer's encouraging and positive comments. The reviewer's specific comments are addressed pointwise below and changes to the text will be highlighted in yellow in the revised manuscript.

(1) The GaMD simulations all include exogenous peptides, thus lacking a control where no such peptide is present (and only stalkless CTF). An earlier study (PNAS 2022 Vol. 119 No. 19 e2113786119) covered this already, but it should be mentioned here that there was no observation of close/activation for the stalkless CTF.

We appreciate the reviewer's concern about the lack of a control where no exogenous peptide is present. As suggested by the reviewer, we are adding more details about the study on the stalkless CTF as a control in the Introduction of the revised manuscript.

(2) Although 5 independent trajectories were generated for each peptide, the authors did not provide sufficient details regarding the convergence of the simulation. This leaves some uncertainties in their results. Given that the binding poses changed relative to the starting docked poses for all three peptides, it is possible that some other binding pockets and/or poses were not explored.

We appreciate the reviewer's comment regarding the convergence of the simulation results. This is clarified in the revised manuscript as:

"We have calculated free energy profiles of individual simulations for each system, including the p9, p17, and p21, as shown below (Figs. S5, S6 and S8). For the p9 peptide, the "Bound" lowenergy state was consistently identified in the 2D free energy profile of each individual simulation (Fig. S5). For the p17 peptide, Pep-GaMD simulations were able to refine the peptide conformation from the "Unbound" to the "Intermediate" and "Bound" states in Sim1 and Sim5, while the peptide reached only the "Intermediate" state in the other three simulations (Fig. S6). For the p21 peptide, Pep-GaMD was able to refine the peptide docking conformation to the

"Bound" state in all the five individual simulations (Fig. S8)."

"It is important to note that the free energy profiles calculated from GaMD simulations of PC1 CTF were not fully converged since certain variations were observed among the individual simulations. Nevertheless, these calculations allowed us to identify representative low-energy binding conformations of the peptides."

(3) The free energy profiles (Figures 2 to 4) based on the selected coordinates provide important information regarding binding and CTF conformational change. However, it is a coarsegrained representation and complementary analysis such as RDFs, and/or contact maps between the peptide and CTF residues might be helpful to understand the details of their interactions. These details are currently only available in the text.

Following the reviewer's suggestion, we have now included a set of protein contact maps showing contacts between the peptides and the TOP domain for each peptide in the representative "Bound" state in revised Supplementary Information (Fig. S4). The contact maps serve to visualize the list of contacts mentioned in the main text. This will be clarified in the revised manuscript.

(4) The use of a stalkless CTF is necessary for studying the functions of the exogenous peptides. However, the biological relevance of the stalkless CTF to ADPKD was not clearly explained, if any.

We appreciate the reviewer's comment. As correctly assessed by the reviewer, the stalkless CTF is not a biological form of PC1 observed in ADPKD, but rather was used as the simplest or least complex system in which the activities and binding of exogenous peptides could be studied. However, in ADPKD, there are numerous missense mutations reported within the GPCR autoproteolysis-inducing (GAIN) domain that have been shown to prevent or inhibit cleavage at the GPCR-coupled proteolysis site (GPS). Loss of PC1 GPS cleavage, which is known to cause ADPKD, would retain or sequester the stalk tethered agonist within the interior of the GAIN domain, which would presumably interfere with interactions between stalk tethered agonist residues and the remainder of the CTF. Furthermore, there are 10 single nucleotide polymorphisms reported within the stalk sequence (ADPKD Variant Database; <https://pkdb.mayo.edu/welcome>), most of which we have found to significantly reduce CTF-mediated activation of the NFAT reporter (Magenheimer BS, et al., [Constitutive signaling by the C-terminal fragment of polycystin1 is mediated by a tethered peptide agonist](#); bioRxiv

2021.08.05.455255). In particular, the ADPKD-associated G3052R stalk mutation that was analyzed along with the stalkless CTF by GaMD simulations (Pawnikar et al, PNAS, 2022) has the same reduction in activity as the stalkless CTF in the cellular signaling reporter assays and the same loss of closed conformation interactions in GaMD analyses. As such, we believe the stalkless CTF has biological relevance from the aspect that it mimics the deficiency in signaling activation observed for PC1 CTF stalk mutants. This is clarified in the revised manuscript in the Introduction, page 5, “constructs encoding a stalkless PC1 CTF (a nonbiological mutant of PC1 with deletion of the first 21 N-terminal residues of CTF) and three ADPKD-associated...” and near the beginning of the Discussion, page 16, where the biological relevance of studying the stalkless CTF is explained

(5) The authors might want to clarify if a stalkless CTF is commonly seen in ADPKD, or if it is just a construct used for this study.

The stalkless CTF is not a biological form of PC1, but rather a construct used for this study. This was clarified in the revised manuscript (see response above).

(6) (Pages 7-8) "...we generated expression constructs of mouse (m) PC1 consisting of the CD5 signal peptide sequence fused in frame with the stalk sequence of mCTF ...". What is the CD5 signal peptide sequence here? What is its use?

The CD5 signal peptide sequence is “MPMGSLQPLATLYLLGMLVASVLG” from the T cell surface glycoprotein, CD5. Since the N-terminus of PC1 CTF is derived from a posttranslational, autocatalytic, endoproteolytic cleavage event, this isoform is already membraneembedded and therefore lacks its endogenous signal peptide. The CD5 signal peptide coding sequence is added to the PC1 CTF expression constructs in order to ensure translation and insertion of the encoded protein at the endoplasmic reticulum. Additional details were added to the Experimental Procedures, page 2 of Supporting Information.

(7) (Page 8) "All peptides were appended with a C-terminal, 7-residue hydrophilic sequence (GGKKKKK) to increase solubility". How did the authors make sure that this sequence has no influence on the signaling?

To determine the possible effect of the hydrophilic GGKKKKK sequence on signaling, we had a ‘solubility tag’ peptide (LGGKKKKK) synthesized and purified by GenScript. It was necessary to add an N-terminal Leu residue to the 7-residue hydrophilic tag sequence in order for the highly hydrophilic peptide to be recovered. Effect of treatment with the solubility tag peptide on activation of the NFAT reporter was assessed for both empty vector- and Δ stalkCTF-transfected cells in 3 separate signaling experiments (see figure below). Each experiment also included a negative control treatment (no peptide/culture medium only addition) and a positive control treatment (stalk peptide p17). The p17 peptide we had available was derived from the stalk sequence of human PC1 that differs from the mouse PC1 sequence at residues 15 and 17, which are two poorly conserved positions within the stalk sequence (see Reviewer 2, Response 3). In the first experiment with the solubility tag and human p17 peptides (B in figure below), we inadvertently used the empty expression vector and Δ stalkCTF expression construct from mouse PC1. After realizing our error, we then performed 2 additional signaling experiments (C and D in figure below) with the ‘correct’ human Δ stalkCTF expression construct and empty vector. In the revised manuscript, we have provided the results from each of the 3 experiments as Fig. S2 (below).

(8) (Page 9) "Using a computational model of the Δ Stalk PC1 CTF developed previously". The authors might want to expand here a little to give a short review about the structure preparation.

We appreciate the reviewer's suggestion regarding the addition of details for structure preparation for Stalkless CTF. We have added these details in section "Docking and Pep-GaMD simulations of peptide agonist binding to stalkless PC1 CTF" on Page 10 in the revised manuscript: "The cryo-EM structure of human PC1-PC2 complex (PDB: 6A70) was used to build the computational model for WT PC1 CTF. As the protein had several missing regions including the Stalk and several loops, homology modeling of the missing regions was done using I-TASSER web server. Using the WT PC1 CTF model, computational model for Δ Stalk was generated by deleting the first 21 residues (3049-3069) of the WT PC1 and using the structure for stalkless CTF, we successfully docked the p9, p17 and p21 stalk peptides with HPEPDOCK. The peptides all bound to the TOP domain and the interface between the TOP domain and extracellular loop 1 (ECL1) of CTF."

(9) How was "contact" defined when counting the number of contacts used in the 2D PMFs (Figures 2-4). Response: We appreciate the reviewer's comment regarding the definition of the number of contacts used in the 2D PMFs. This has been clarified in the revised manuscript as: "The number of contacts is calculated between any atom pairs within 4 Å distance of the peptide and extracellular domains of PC1 protein."

(10) How was the ranking of GaMD clusters done? It looks from Figure 3A that the "intermediate" state is more favorable compared to the "bound" state, but it was claimed in the text the "bound" state was ranked 1st.

Thanks to the reviewer for this comment. It has been clarified in the revised

Supplementary Information: "Three independent Pep-GaMD simulations were combined to perform structural clustering using the hierarchical agglomerative clustering algorithm in CPPTRAJ. A 3 Å RMSD cutoff was used for each peptide system. PyReweighting was then applied to calculate the original free energy values of each peptide structural cluster with a cutoff of 500 frames. The structural clusters were finally ranked according to the reweighted free energy values." And in the revised main text: "It is important to note that the free energy profiles calculated from GaMD simulations of PC1 CTF were not fully converged since certain variations were observed among the individual simulations. The free energy values of 2D PMF minima shown in Figure 3A could differ from those in the 1D PMF minima of peptide structural clusters, especially with the usage of distinct reaction coordinates. Nevertheless, these calculations allowed us to identify representative low-energy binding conformations of the peptides."

(11) When mentioning residue pair distances, such as in the sentence "The distance between the TOP domain residue R3848 and PL residue E4078 was 3.8 Å (Fig. 4D)" on page 12, it should be clarified if these distances are average distance, or a statistical error can be given.

We appreciate the reviewer's comment regarding the TOP Domain and PL distance between residues R3848-E4078. This has been clarified on page 14 in the revised manuscript as:

"The distance between the TOP domain residue R3848 and PL residue E4078 was 3.8 Å. The distance was extracted from the top-ranked structural cluster of the p21 bound to the Δ Stalk CTF, corresponding to the "Closed/Active" low-energy conformational state. (Fig. 4E)".

(12) More analysis of the GaMD can be performed. For example, the authors observed a single "bound" state for p21, but there must be some flexibility in the peptide and the protein itself. The authors might want to consider adding some plots illustrating the flexibility of the peptide residues (for example, a RMSD plot). Contact maps can also be added to visualize the results currently discussed in the text.

We thank the reviewer for their constructive suggestions. To characterize flexibility of the peptide and protein in the revised manuscript, we have added plots of the TOP-PL interaction distance between residues R3848-E4078 in PC1, the radius of gyration (Rg) of p21 and root-mean square deviation (RMSD) of p21 relative to the starting HPEPDOCK conformation of the peptide in the new Fig. S7. The peptide-protein contact map has also been added in the new Fig. S4.

(13) (Page 7) In the sentence "...sampled the "Closed/Active" low-energy state relative to the large number of Stalk-TOP contacts," I suggest using "related to" instead of "relative to"

We thank the reviewer for the comment, and we have replaced "relative to" to "related to" in the following sentence "...sampled the "Closed/Active" low-energy state relative to the large number of Stalk-TOP contacts"

(14) (Page 7) In the sentence Our previous study utilized expression constructs of human PC1 CTF, however, in order to prepare for ..., "PC1 CTF, however," -> "PC1 CTF. However,"

We thank the reviewer for the comment, and we have replaced "PC1 CTF, however," to "PC1 CTF. However," in the following sentence Our previous study utilized expression constructs of human PC1 CTF, however, in order to prepare for

Reviewer 2:

The autosomal dominant polycystic kidney disease (ADPKD) is a major form of polycystic kidney disease (PKD). To provide better treatment and avoid side effects associated with currently available options, the authors investigated an interesting GPCR, polycystin-1 (PC1), as a potential therapeutic target. In vitro and in silico studies were combined to identify peptide agonists for PC1 and to elucidate their roles in PC1 signaling. Overall, regarding the significance of the findings, this work described valuable peptide agonists for PC1 and the combined in vitro and in silico approach can be useful to study a complex system like PC1. However, the strength of the evidence is incomplete, as more experiments are needed as controls to validate the computational observations. The work appears premature.

We greatly appreciate the reviewer's encouraging and positive comments. The reviewer's specific comments are addressed pointwise below and changes to the text will be highlighted in yellow in the revised manuscript.

(1) *The therapeutic potential of PC1 peptide agonists is unclear in the introduction. For example, while the FDA-approved drug Jynarque was mentioned, the text was misleading as it sounded like Jynarque targeted PC1. In fact, it targets another GPCR, the vasopressin receptor 2 (V2). A clear comparison of targeting PC1 over V2 pathways and their therapeutic relevance can help the readers better understand the importance of this work. Importantly, a clear background on the relationship between PC1 agonism and treatments for ADPKD is necessary.*

We understand the confusion that was caused by the brevity of our introductory paragraph and will clarify the differences in therapeutic targeting between Jynarque and our PC1 stalk-derived peptides in the revised manuscript. We will also expound on the rationale for targeting PC1 agonism as a therapeutic approach for ADPKD versus Jynarque. For example: It is known that ADPKD disease severity is dependent on the functional levels of PC1. Jynarque is a small molecule antagonist of the arginine vasopressin receptor 2, V2R, whose signaling, and production of cAMP has been shown to be increased in ADPKD. As this drug targets one

of the downstream aberrant pathways, it is only capable of slowing disease progression and has numerous undesirable side effects. We reasoned that a therapeutic agent capable of stimulating and thus augmenting PC1 signaling function would be a safer, cyst initiation-proximal treatment capable of preventing cyst formation with few side effects.

(2) PC1 is a complex membrane protein, and most figures focus on the peptide-binding site. For general readers (or readers that did not read the previous PNAS publication), it is hard to imagine the overall structure and understand where the key interactions (e.g., R3848-E4078) are in the protein and how peptide binding affects locally and globally. I suggest enhancing the illustrations.

We thank the reviewer for the constructive comment on adding more illustrations for the PC1 protein to understand the overall structure and the location of the key interaction R3848E4078. We have included these suggestions and modified the main figures in the revised manuscript.

(3) The authors used the mouse construct for the cellular assays and the peptide designs in preparation for future in vivo assays. This is helpful in understanding biology, but the relevance of drug discovery is weakened. Related to Point 1, the therapeutic potential of PC1 peptide agonist is largely missing.

The therapeutic potential of a PC1 peptide agonist is addressed in response #1 above. As mentioned in the manuscript and recognized by the reviewer, the cellular signaling assays were performed with the mouse PC1 CTF expression construct and with peptides based on the mouse PC1 stalk sequence for future, pre-clinical studies, while the peptide binding studies were performed with the human PC1 stalk sequence. We feel the relevance for drug discovery is not significantly weakened for a number of reasons: 1) as shown in Fig. 1A, the stalk sequence is highly conserved between mouse and human PC1, specifically there are only 2 residue differences present within peptides p17 and p21. One of the differences is a ‘semi-conservative’ Gln-Arg substitution at peptide residue 15, while the second difference is a conservative Ile-Val substitution at peptide residue 17; 2) we have found that an Arg to Cys mutation within the mouse PC1 CTF stalk has the same effect on signaling as the corresponding human Gln to Cys ADPKD-associated mutation which was analyzed in Pawnikar et al., 2022; and 3) both peptide residues 15 and 17 represent highly variable positions within the PC1 stalk as shown in the sequence logo (below) of the stalk sequence from 16 vertebrate species; and 4) while addressing the potential effect of the hydrophilic solubility tag on stalk peptide-mediated rescue of CTF Δ stalk signaling (see Reviewer 1 comments, point #7), we utilized the ‘human’ version of p17 as a positive control and tested its activation with both mouse and human CTF Δ stalk expression constructs and found that human p17 peptide was also capable of stimulating the mouse CTF Δ stalk protein (Fig. S2).

Author response image 1.



(4) More control experiments are needed. For example, a 7-residue hydrophilic sequence (GGKKKKK) is attached to the peptide design to increase solubility. This 7-residue peptide should be tested for PC1 activation as a control. Second, there is no justification for why the peptide design must begin with residue T3041. Can other segments of the stalk also be agonists?

As mentioned above for Reviewer 1, the hydrophilic peptide has been synthesized and tested for activation of signaling by the stalkless CTF in the revised manuscript as Fig. S2. The design of peptides that begin with residue T3041 of mouse PC1 CTF is modeled on numerous similar studies for the family of adhesion GPCRs. Optimization of the binding and activity of the PC1 peptide agonist will be investigated in future studies and could include such parameters as whether the peptide must include the first residue and whether subsegments of the stalk are also agonists, however, we feel these questions are beyond the scope of this initial report.

(5) There are some major concerns about the simulations: The GaMD simulations showed different binding sites of p-21, p-17, and p-9, and the results report the simulated conformations as "active conformational states". However, these are only computational findings without structural biology or mutagenesis data to validate. Further, neither docking nor the simulation data can explain the peptide SAR. Finally, it will be interesting if the authors can use docking or GaMD and explain why some peptide designs (like P11-P15) are less active (as control simulations).

The reviewer brings up an important observation regarding differences in binding sites between peptides p9, p17 and p21. We will include discussion of this observation and our interpretations to the revised manuscript. While the present study is focused on identification of initial peptides that are able to activate the PC1 CTF, we shall include further mutation experiments and simulations, peptide SAR and optimization of the lead peptides in future studies. This has been clarified in the revised manuscript.

(6) Additional experiments for the controls and for validating the simulations. Additional simulations to explain the SAR.

We appreciate the reviewer's comment for additional experiments for the controls and additional simulations to explain the SAR. For future studies, we shall include further mutation experiments and simulations, peptide SAR and optimization of the lead peptides.

(7) What is the selectivity of the peptides between PC1 and PC2?

We have not tested the selectivity of the peptides for PC1 versus PC2 primarily because transfection of PC2 does not activate the NFAT reporter. However, it is possible that co-transfection of PC2 with the PC1 CTF could alter stalk peptide binding. This will be important to consider in future studies.

Reviewer 3:

The authors demonstrate the activation of Polycystin-1 (PC1), a G-protein coupled receptor, using small peptides derived from its original agonist, the stalk TA protein. In the experimental part of the study, the authors performed cellular assays to check the peptide-induced reactivation of a mutant form of PC1 which does not contain the stalk agonist. The experimental data is supported by computational studies using state-of-the-art Gaussian accelerated Molecular Dynamics (GaMD) and bioinformatics analysis based on sequence covariance. The computer simulations revealed the mechanistic details of the binding of the said peptides with the mutant PC1 protein and discovered different bound, unbound, and intermediate conformations depending on the peptide size and sequence. The use of reliable and well-established molecular simulation algorithms and the physiological relevance of this protein autosomal dominant polycystic kidney disease (ADPKD) make this work particularly valuable.

We greatly appreciate the reviewer's encouraging and positive comments. The reviewer's specific comments are addressed pointwise below and changes to the text will be highlighted in yellow in the revised manuscript.

(1) No control has been used for the computational (GaMD) study as the authors only report the free energy surface for 3 highly agonistic peptides but for none of the other peptides that did not induce an agonistic effect. Therefore, in the current version, the reliability of the computational results is not foolproof.

We appreciate the reviewer's concern about the lack of control with the other peptides that did not induce an agonistic effect. To address the reviewer's concern, we have included more details on the study of the stalkless CTF and the solubility tag peptide (Fig. S2) as controls in the revised manuscript.

(2) All discussions about the residue level interactions focused only on geometric aspects (distance, angle, etc) but not the thermodynamic aspect (e.g. residue-wise interaction energy). Considering they perform a biased simulation; the lack of interaction energy analysis only provides a qualitative picture of the mechanism.

As mentioned by the reviewer, we have added MM/PBSA analysis results in the revised manuscript and SI.

Molecular Mechanics/Poisson-Boltzmann Surface Area (MM/PBSA) analysis was performed to calculate the binding free energies of peptides p9, p17 and p21 to PC1 CTF. The analysis was performed using the trajectory in which the peptide was bound to the receptor. In MM/PBSA, the binding free energy of the ligand (L) to the receptor (R) to form the complex (RL) is calculated as:

$$\Delta G_{bind} = G_{RL} - G_R - G_L \quad (1)$$

where G_{RL} is the Gibbs free energy of the complex RL, G_R is the Gibbs free energy of the molecule R in its unbound state and G_L is the Gibbs free energy of the molecule L in its

unbound state, respectively.

ΔG_{bind} can be divided into contributions of different interactions as:

$$\Delta G_{bind} = \Delta H - T\Delta S = \Delta E_{MM} + \Delta G_{sol} - T\Delta S \quad (2)$$

in which

$$\Delta E_{MM} = \Delta E_{int} + \Delta E_{elec} + \Delta E_{vdW} \quad (3)$$

$$\Delta G_{sol} = \Delta G_{PB/GB} + \Delta G_{SA} \quad (4)$$

$$\Delta G_{SA} = \gamma \cdot SASA + b \quad (5)$$

where ΔE_{MM} , ΔG_{sol} , ΔH and $-T\Delta S$ are the changes in the gas-phase molecular mechanics (MM) energy, solvation free energy, enthalpy and conformational entropy upon ligand binding, respectively. ΔE_{MM} includes the changes in the internal energies ΔE_{int} (bond, angle and dihedral energies), electrostatic energies ΔE_{elec} , and the van der Waals energies ΔE_{vdW} . ΔG_{sol} is the sum of the electrostatic solvation energy $\Delta G_{PB/GB}$ (polar contribution) and the nonpolar contribution ΔG_{SA} between the solute and the continuum solvent. The polar contribution is calculated using either the Poisson Boltzmann (PB) or Generalized Born (GB) model, while the nonpolar energy is usually estimated using the solvent-accessible surface area (SASA) where γ is surface tension coefficient and b is the constant offset. The change in conformational entropy $-T\Delta S$ is usually calculated by normal-mode analysis on a set of conformational snapshots taken from MD simulations. However, due to the large computational cost, changes in the conformational entropy are usually neglected as we were concerned more on relative binding free energies of the similar peptide ligands.

MM/PBSA analysis was performed using the gmx_MMPBSA software with the following command line:

```
gmx_MMPBSA -O -i mmpbsa.in -cs com.tpr -ci index.ndx -cg 1 13 -ct com_traj.xtc -cp topol.top -
o FINAL_RESULTS_MMPBSA.dat -eo FINAL_RESULTS_MMPBSA.csv Input file for running
MM/PBSA analysis:
```

```
&general
```

```
sys_name="Prot-Pep-CHARMM",
```

```
startframe=1, endframe=200, # In gmx_MMPBSA v1.5.0 we have added a new PB radii set
named charmm_radii.
```

```
# This radii set should be used only with systems prepared with CHARMM force fields.
```

```
# Uncomment the line below to use charmm_radii set
```

```
# PBRadii=7,
```

```
&pb
```

```
# radii=0 is recommended which means using radii from the prmtop file for both the PB
calculation and for the NP
```

calculation

istrng=0.15, fillratio=4.0, radiopt=0

The relative rank of the overall peptide binding free energies (Table S1) was consistent with the experimental signaling data, i.e., p21>p9>p17, for which p21 showed the largest binding free energy value of binding (-40.29 ± 6.94 kcal/mol).

(3) It is not mentioned clearly whether the reader should interpret the free energy landscapes quantitatively or qualitatively. Considering no error analysis or convergence plots are reported for the GaMD free energy surfaces, it may be assumed the results are qualitative. The readers should consider this caveat and not try to quantitatively reproduce these free energy landscapes with other comparable techniques.

We appreciate the reviewer's comment whether the free energy landscapes should be interpreted quantitatively or qualitatively. The presented free energy landscapes could be considered semi-quantitative since the simulations are not fully converged. This will be clarified in the revised manuscript as: "It is important to note that the free energy profiles calculated from GaMD simulations of PC1 CTF were not fully converged since certain variations were observed among the individual simulations. Nevertheless, these calculations allowed us to identify representative low-energy binding conformations of the peptides."

(4) Energy decomposition analysis similar to the following paper (<https://pubs.acs.org/doi/10.1021/bi201856m>) should be provided to understand the residue level enthalpic contribution in the peptide-protein interaction.

As mentioned by the reviewer, we have performed residue-wise interaction energy analysis and included the analysis results in the revised manuscript and SI.

Residue-wise interaction energy analysis was performed on peptides p9, p17 and p21 using the trajectory in which the peptide was bound to the PC1 CTF using the gmx_MMPBSA software with the following command line:

```
gmx_MMPBSA -O -i mmpbsa.in -cs com.tpr -ct com_traj.xtc -ci index.ndx -cg 3 4 -cp topol.top -o
FINAL_RESULTS_MMPBSA.dat -eo FINAL_RESULTS_MMPBSA.csv -do
FINAL_DECOMP_MMPBSA.dat -deo FINAL_DECOMP_MMPBSA.csv
```

Input file for running residue-wise energy decomposition analysis:

&general

sys_name="Decomposition", startframe=1, endframe=200,

forcefields="leaprc.protein.ff14SB"

&gb

igb=5, saltcon=0.150,

make sure to include at least one residue from both the receptor #and peptide in the print_res mask of the &decomp section.

this requirement is automatically fulfilled when using the within keyword.

<http://archive.ambermd.org/201308/0075.html>

&decomp

idecomp=2, dec_verbose=3, print_res="A/854-862 A/1-853",

Residue-wise energy decomposition analysis allowed us to identify key residues that contributed the most to the peptide binding energies. These included residues T1 and V9 in p9 (Table S2), residues T1, R15 and V17 in p17 (Table S3), and residues P10, P11, P19 and P21 in p21 and residue W3726 in the PC1 CTF (Table S4). The energetic contributions of these residues apparently correlated to the sequence coevolution predicted from the Potts model.

(5) To showcase the reliability of the computational approach, the authors should perform the MD simulation studies with one peptide that did not show any significant agonistic effect in the experiment. This will work as a control for the computational protocol and will demonstrate the utility of the pep-GaMD simulation in this work.

We appreciate the reviewer's concern about the lack of control with the other peptides that did not induce an agonistic effect. It is difficult for us to add more MD simulations on the other peptides, due to student leave after PhD graduation. But to address the reviewer's concern, we have included more details on the study of the stalkless CTF as a control in the revised manuscript.

(6) To assess the accuracy of the computational results the authors should mention (either in the main text or SI) whether the reported free energy surfaces were the average of the five simulations or computed from one simulation. In the latter case, free energy surfaces computed from the other four simulations should be provided in the SI. In addition, how many binding unbinding events have been observed in each simulation should be mentioned.

We appreciate the reviewer's comment regarding convergence of the simulation free energy surfaces. In response to Reviewer 1, we have calculated free energy profiles of individual simulations for each system, including the p9, p17, and p21 (Figs. S5, S6 and S8).

"We have calculated free energy profiles of individual simulations for each system, including the p9, p17, and p21 (Figs. S5, S6 and S8). For the p9 peptide, the "Bound" low-energy state was consistently identified in the 2D free energy profile of each individual simulation (Fig. S5). For the p17 peptide, Pep-GaMD simulations were able to refine the peptide conformation from the "Unbound" to the "Intermediate" and "Bound" states in Sim1 and Sim5, while the peptide reached only the "Intermediate" state in the other three simulations (Fig. S6). For the p21 peptide, PepGaMD was able to refine the peptide docking conformation to the "Bound" state in all the five individual simulations (Fig. S8)."

"It is important to note that the free energy profiles calculated from GaMD simulations of PC1 CTF were not fully converged since certain variations were observed among the individual simulations. Nevertheless, these calculations allowed us to identify representative low-energy binding conformations of the peptides."

<https://doi.org/10.7554/eLife.95992.2.sa0>